



Stealthy shorts: Informed liquidity supply[☆]

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ABSTRACT

Short sellers are widely known to be informed, which would typically suggest that they demand liquidity. We obtain comprehensive transaction-level data to decompose daily short volume into liquidity-demanding and liquidity-supplying components. Contrary to conventional wisdom, we show that the most informed short sellers are actually liquidity suppliers, not liquidity demanders. They are particularly informative about future returns on news days and trade on prominent cross-sectional return anomalies. Our analysis suggests that market making and opportunistic risk-bearing are unlikely to explain these findings. Instead, our results align with recent market microstructure theory, pointing to strategic liquidity provision by informed traders.

1. Introduction

There is a long-standing paradox in the short selling literature: Short sellers, despite being widely recognized as informed investors, are also associated with improved market liquidity.¹ This puzzle arises because, in the standard microstructure literature, market makers react to the presence of informed trading by widening bid-ask spreads to protect against adverse selection, thereby reducing liquidity (e.g., [Glosten and Milgrom, 1985](#) and [Kyle, 1985](#)). This raises an important question: How can short sellers, as a whole, simultaneously be informed and contribute positively to liquidity?

The natural explanation for this paradox is that there are two types of short sellers. One type comprises the informed short sellers who use market orders and demand liquidity to act quickly on their informational advantage. The other type comprises (voluntary) market makers who place uninformed limit orders, thereby supplying liquidity ([Boehmer et al., 2008](#) and [Comerton-Forde et al., 2016](#)).

In this paper, we shed light on the informational advantage of short sellers and examine which short trades are more informative about future returns: those that demand liquidity or those that supply it. Surprisingly, we find that liquidity-supplying short sellers are actually more informed than liquidity-demanding short sellers, a result that contradicts much of the existing literature in this area.

One of the main challenges in separating short sellers into liquidity demanders and liquidity suppliers is the limited granularity of commonly used short sale data, which are available at the stock level on a daily basis and do not distinguish between different motives driving short seller behavior. These aggregated measures potentially mask the heterogeneous effects of different types of short sales on return predictability. To overcome this challenge, we (1) obtain comprehensive transaction-level short sale data from each of the eleven main U.S. equity exchanges over a five-year period from 2014 through 2018 and (2) employ a three-step matching algorithm to link each short sale trade to the prevailing market quotes. By using trade and quote prices

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¹ There is substantial evidence that short sellers are informed traders. High levels of short selling activity are associated with predictably low future returns and improved price efficiency (see, e.g. [Asquith et al., 2005](#); [Boehmer et al., 2008](#); [Diether et al., 2009](#); [Rapach et al., 2016](#); and [Saffi and Sigurdsson, 2011](#)). At the same time, short selling is also positively related to market liquidity, with restrictions on short sales generally resulting in decreased liquidity (see, e.g., [Beber and Pagano, 2013](#) and [Bris et al., 2007](#)).

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and the Lee and Ready (1991) classification algorithm, we decompose the short sale volume based on whether the short seller supplies or demands liquidity and analyze the potentially different predictability of returns from these two components. This analysis provides insights into which type of short selling is most informed and at which time horizon. By disaggregating the standard daily short sale measure, we can better understand the trading strategies of short sellers and how their informational advantage contributes to the efficient incorporation of information into stock prices.

We begin our analysis by replicating a well-documented pattern in the cross-section of stock returns: stocks with a high short sale volume to total trading volume ratio underperform those with a low short volume ratio (see, e.g., Boehmer et al., 2008; Diether et al., 2009; and Rapach et al., 2016). More importantly, when we decompose this ratio into liquidity-supplying short volume ratio (*LSS*) and liquidity-demanding short volume ratio (*LDS*), we uncover interesting heterogeneous patterns. Contrary to conventional wisdom, our portfolio analysis shows that only *LSS* negatively predicts future equity returns. In particular, stocks in the highest *LSS* quintile underperform those in the lowest quintile by a risk-adjusted return of 38 basis points (bps) over a 21-day holding period. In contrast, the predictive power of *LDS* over the same holding period is much weaker, at just 12 bps, which is statistically indistinguishable from zero and driven entirely by the return on the day after portfolio formation. In cross-sectional regressions, we show that the return predictability associated with *LSS* is not subsumed when controlling for other well-known short selling metrics and standard return predictors. This suggests that liquidity-supplying short sales contain unique information about future stock returns. In additional tests, we find that our documented predictability is neither concentrated in stocks with specific characteristics, nor driven by particular time periods or samples of stocks, and it holds for various alternative holding periods. Overall, our results indicate strong predictability of future returns from liquidity-supplying short sales, while such predictability is absent for liquidity-demanding short sales over holding periods longer than one day. These findings suggest that liquidity-supplying short sales may represent informed trading by short sellers with relatively long-lived information.

In addition to informed trading, the return predictability associated with liquidity-supplying short sales might partly reflect compensation for voluntary market making or opportunistic risk bearing (Diether et al., 2009). To explore this possibility, we use a decomposition method similar to that of Boehmer et al. (2021b) to separate the variation in *LSS* into proxies for these two components and a residual component. Our findings show that only the residual component drives the predictive power of *LSS* for future returns, while the market-making and risk-bearing components do not predict cross-sectional stock returns in a statistically significant way.

We shed further light on the nature of return predictability and its link to the informational advantage inherent in liquidity-supplying short sales. First, we use a framework similar to that of Engelberg et al. (2012), and examine the predictive power of *LSS* for stock returns around firm-specific news release days. If the predictive ability of liquidity-supplying short sales stems from an informational advantage, then *LSS* should predict returns particularly on those days when news is released and information is incorporated into stock prices. Consistent with this hypothesis, we find that liquidity-supplying short sales are particularly informative about future returns on news days compared to non-news days. Additionally, as with our decomposition method, we demonstrate that this increased predictive power is present exclusively in the residual component of *LSS*, further supporting the notion that liquidity-supplying shorts are, on average, informed about company fundamentals.

As a second empirical exercise, we study whether liquidity-supplying short sellers use their informational advantage and trade on long-term information embedded in cross-sectional equity anomalies. We follow Wang et al. (2020) in analyzing over- and undervaluation

ranks of stocks in portfolios sorted on short volume ratio. We find that liquidity-supplying short sellers react to information embedded in anomalies by shorting overvalued stocks more than undervalued ones. The same is not true for liquidity-demanding shorting flows. Overall, our findings indicate that informed short sellers capitalize on longer-term price decreases by supplying liquidity rather than taking it out of the market (see, e.g., Kaniel and Liu, 2006 and Riccò et al., 2024).

To corroborate that liquidity-supplying short sellers trade on long-term information, we examine whether the return predictability of liquidity-supplying short sales is concentrated on trading venues that favor patient trading strategies. In particular, we exploit heterogeneity in exchange fee structures and find that the predictive power of *LSS* is driven almost entirely by trades executed on maker-taker venues—venues that reward passive limit order submission and are more attractive to traders acting on long-lived information. In contrast, *LSS* on inverted venues, which incentivize more aggressive limit order placement, is not predictive of longer-run returns. These findings are consistent with the theoretical predictions in Malinova and Park (2015) and Battalio et al. (2016), who emphasize how exchange fee structures influence the behavior of informed traders.

Our findings contribute to several important strands of literature. First, we add to the extensive body of research on the role of short sellers as informed traders. While previous studies have documented that short selling activity predicts lower future returns and reflects valuable information about firm fundamentals (see, e.g., Aitken et al., 1998; Asquith et al., 2005; Boehmer et al., 2008; Christophe et al., 2010; Dechow et al., 2001; Desai et al., 2002; Deshmukh et al., 2015; Diether et al., 2009; Engelberg et al., 2025; and Rapach et al., 2016), these studies typically rely on aggregated short sale measures, which mask the different strategies short sellers employ to leverage their informational advantage. By disaggregating short sale activity into liquidity-supplying and liquidity-demanding components, our paper advances the understanding of how informed short sellers operate and how their trades contribute to the efficient incorporation of information into stock prices. This distinction is crucial for evaluating whether informed short sellers are inherently detrimental to market liquidity or if they can simultaneously enhance liquidity through their trading activities.

The study closest to ours is Comerton-Forde et al. (2016), who also differentiate between short sales that provide liquidity and those that demand it. Since these authors find that liquidity-demanding shorts are followed by lower returns, the clever reader might notice that our results seem at odds with theirs. Appendix C provides a reconciliation of our results with those of Comerton-Forde et al. (2016). The key distinction is that their work focuses on intraday trading, whereas we study longer-term return predictability. Our results suggest that informed short sellers capitalize on longer-term information by supplying liquidity. In general, our findings are consistent with the idea that informed short sellers play a key role in price efficiency, without necessarily compromising market liquidity.

Second, our research engages with the microstructure literature on the interaction between informed trading and liquidity provision. Traditional models, such as those proposed by Glosten (1994), Glosten and Milgrom (1985) and Kyle (1985), suggest that informed traders typically demand liquidity, which contrasts with our findings. Our findings align with more recent theoretical work (see, e.g., Brolley and Malinova, 2021; Chakravarty and Holden, 1995; Goettler et al., 2009; Hollifield et al., 2004; Kaniel and Liu, 2006; Riccò et al., 2024; Roşu, 2020; and Seppi and Kumar, 1994), which posits that informed traders may strategically supply liquidity, particularly when their information is long-lived and there is no immediate need for trade execution. Similarly, this perspective is supported by a handful of recent empirical studies that show informed traders, such as insiders, often use limit orders or time their trades to minimize market impact and avoid revealing their informational advantage (see, e.g., Ahern, 2020; Akey et al., 2022; Collin-Dufresne and Fos, 2015; and Kacperczyk and Pagnotta,

2019). Our work broadens the scope of these studies, which typically focus on specific events or narrow subsets of trades, by studying short sales that account for approximately 45% of total trading volume. More importantly, we demonstrate that *only* liquidity-supplying short sales predict longer-term returns, while liquidity-demanding short sales do not, thereby highlighting the distinct role of the former in driving price efficiency.

The remainder of the paper is organized as follows. Section 2 details the data sources, describes our methodology for decomposing short volume into liquidity-supplying and liquidity-demanding components, and provides summary statistics. Section 3 presents our main empirical findings. Section 4 explores the underlying mechanisms behind the results documented in Section 3. Section 5 concludes.

2. Data

This section introduces our sample and provides descriptive statistics. Section 2.1 explains our sample construction of intraday time-stamped short sale transactions reported on U.S. equity exchanges. Section 2.2 details the methodology used to classify short sales as either seller-initiated or buyer-initiated. Finally, Section 2.3 presents the corresponding descriptive statistics.

2.1. Sample construction

We use the NYSE Trade and Quote (TAQ) dataset, which provides comprehensive transaction level data for trades and quotes on the U.S. equity market. However, the TAQ data do not include flags to identify short sales, presenting a significant limitation for understanding short selling activities at the intraday level. To address this gap, we obtain proprietary short sale information at the transaction level from each of the three largest exchange groups in the U.S., viz. the NYSE, Nasdaq, and Cboe (which acquired BATS Global Markets in 2017). By the end of our sample period, the eleven exchanges comprising the NYSE, Nasdaq, and Cboe collectively accounted for approximately 97% of the exchange trading volume in U.S. equity markets.² The short selling data include the ticker symbol, exchange code, trade price, trade size, sales conditions (i.e., exempt or non-exempt from the price test of SEC's Regulation SHO, Rule 201), date, and a timestamp at second or millisecond level. We use these variables to match every single short sale transaction with the corresponding trade in the TAQ data. Following Hu et al. (2025), we include non-exempt short sales during regular market hours (9:30 am to 4:00 pm).³ These data allow us to identify transactions of opening short sale trades, but we do not observe when short positions are reversed or closed. Our dataset covers a sample period from 2014 through 2018.⁴

² The remaining trading volume takes place at the Investors Exchange (IEX), which was approved by the SEC as an official exchange in mid-2016. More detailed trade volume information for each exchange is available at https://www.cboe.com/us/equities/market_share/market. The eleven exchanges, categorized by group, include the following: for the NYSE Group – NYSE, NYSE American, NYSE Arca, and NYSE National; for the Nasdaq Group – Nasdaq, Nasdaq BX, and Nasdaq PSX; and for the Cboe – BZX, BYX, EDGA, and EDGX. While NYSE Chicago has also been part of the NYSE Group since July 2018, historical short sale trade data for this venue is only available from November 4, 2019, and thus outside our sample period. Further details on the data availability can be found at: https://www.nyse.com/publicdocs/nyse/data/Monthly_Short_Sales_Client_Spec_v1.3.pdf

³ Under SEC Regulation SHO, Rule 201, certain short sale trades are exempt from the price test restrictions. These exemptions include trades related to bona fide market-making, riskless principal transactions, and specific cases like odd-lot and volume-weighted average price transactions. By excluding exempt short trades, we aim to better capture the trading of potentially informed short sellers. Further details on Regulation SHO, Rule 201, can be found at: <https://www.sec.gov/files/rules/final/2010/34-61595-secg.htm>

We complement our transaction-level data with daily stock returns, share prices, shares outstanding, and other stock characteristics from the Center for Research in Security Prices (CRSP). We merge the short volume data with the CRSP database by mapping ticker symbols to PERMNO identifiers. Additionally, to ensure that our findings are distinct from those using other short sale metrics previously introduced in the literature, we incorporate two robust predictors of cross-sectional stock returns as control variables in our analyses: active utilization, measured as the ratio of shares on loan to shares available for loans (Boehmer et al., 2021a and Saffi and Sigurdsson, 2011), and the shorting fee (Drechsler and Drechsler, 2021 and Engelberg et al., 2025) from IHS Markit. We also account for different stock-day measures of liquidity from Intraday Indicators (IID) by Wharton Research Data Services (WRDS).⁵ Furthermore, we collect stock-level accounting information from Compustat and the Open Source Asset Pricing database of Chen and Zimmermann (2022). Lastly, we merge our stock-level short volume information with news release information from the RavenPack Analytics (RPA) database. Our sample includes common stocks (share codes 10 and 11) listed on the NYSE, Amex, and Nasdaq (exchange codes 1, 2, and 3). We exclude stocks with non-positive book-to-market ratios, stocks below the 10th NYSE market capitalization percentile and stocks with a price below \$1 at the end of the previous month. These sample restrictions help alleviate concerns that our results are driven by illiquid microcaps.⁶ More details on variable definitions are provided in Appendix A.

2.2. Identification of liquidity-supplying and liquidity-demanding short sales

The objective of our paper is to examine the dynamics and the informational content of two distinct types of short sales: liquidity-supplying short sales and liquidity-demanding short sales. To this end, we first match each short sale transaction to the corresponding observation in the TAQ data, using all transaction details observable in both datasets. In particular, we employ the WRDS Consolidated Trades (WCT) database, which contains not only all equity trades but also the best available national bid and offer (NBBO) prices at the time of each transaction.⁷ We employ a three-step matching procedure to link every short sale transaction to a TAQ trade and to determine the prevailing NBBO quote, as detailed in Appendix B.

The classification of each short sale transaction relies on inferring the trade direction by comparing the trade price of the short sale to the NBBO quoted prices at the intraday level. We apply the Lee and Ready (1991) algorithm, which integrates both the quote and the tick tests to determine the direction for each trade at the prevailing midpoint quote. Short sale transactions below the prevailing midpoint quote are categorized as short seller-initiated, while those above it are classified as buyer-initiated. For the remaining short trades executed at the midquote, we apply the tick test and classify a short trade as seller-initiated (buyer-initiated) if the trade price is below (above) the preceding trade price. Using INET order data and a subsample of Nasdaq stocks, Chakrabarty et al. (2012) compare the true percentage of

⁴ Note that the transaction-level short sales are published in monthly files with a publication lag of around two weeks. As such, market participants are not able to observe and trade on these data immediately. Hence, the observed predictability in this paper should be interpreted as a measure of informational content of short sales rather than as real-time information for a trading strategy.

⁵ WRDS IID provides stock-level variables at daily frequency such as liquidity spreads, intraday volatility, Kyle's Lambda, and market open-to-close measures of volume and return, calculated using NYSE TAQ data.

⁶ Our main results remain robust across various adjustments in sample construction.

⁷ Based on correspondence with WRDS, the WCT dataset contains the complete NBBO, as suggested by Holden and Jacobsen (2014). The WCT database is also used in, e.g., Barber et al. (2024).

short sales that are either seller- or buyer-initiated to the corresponding percentages based on the Lee and Ready (1991) algorithm. Importantly for our analysis, they find that misclassification rates for short sales are indistinguishable from zero at the daily stock level. Throughout this paper, we will refer to buyer-initiated short sales as liquidity-supplying and seller-initiated short sales as liquidity-demanding.

In the final step, we aggregate short sale transactions into three daily stock-level measures of short selling activity. In line with the literature (see, e.g., Boehmer et al., 2008; Diether et al., 2009; and Wang et al., 2020), we define *ShortingFlow* as the daily number of shares sold short divided by the total number of shares traded for a stock during the same day. In addition, we introduce two novel metrics critical to our analysis: the liquidity-supplying short volume ratio (*LSS*) and the liquidity-demanding short volume ratio (*LDS*). *LSS* is calculated as the daily number of liquidity-supplying shares sold short for a stock divided by its daily total trading volume, whereas *LDS* represents the daily number of liquidity-demanding shares sold short for a stock divided by its total trading volume. The total trading volume is the total daily trading volume on the three major U.S. exchanges in our sample.

2.3. Summary statistics

Panel A of Table 1 presents daily stock-level summary statistics of the short sale data and other stock characteristics. We present the time-series averages of the cross-sectional mean, standard deviation, 10th percentile, median, and 90th percentile of the variables.

On average, short selling represents 45.5% of total trading volume in our sample. This magnitude of shorting flow is significantly larger compared to that in earlier samples (see, e.g., Diether et al., 2009) but comparable to estimates reported in recent studies (see, e.g., Reed et al., 2020 and Wang et al., 2020). After we apply the trade classification algorithm to our short sale data, we find that liquidity-supplying (21.4%) and liquidity-demanding (22.5%) shorts contribute almost equally to the overall trading volume.⁸ Studies that use more limited and less recent data report somewhat larger shares of liquidity-supplying short sales (see, e.g., Chakrabarty et al., 2012 and Comerton-Forde et al., 2016). For instance, for a sample of 200 Nasdaq-listed stocks during two months in 2005, Chakrabarty et al. (2012) find that 43.3% of short sales are seller-initiated or liquidity-demanding. This difference decreases for a sample in 2008, in which the share of short sales executed at the ask price (39.8%) is similar to that executed at the bid price (35.9%) (Comerton-Forde et al., 2016). Overall, recent data indicate that short sales have gained significant importance over the years, with both buyer- and seller-initiated shorts being equally important.

In Fig. 1 we plot the five-day moving average of short selling volume as a percentage of total volume over time. Additionally, we break down this ratio by the three major exchange groups (NYSE, Nasdaq, and Cboe) and the two types of short sales, as determined by our classification algorithm. We observe that shorting flow varies over time, ranging between approximately 37% and 49%. There is significant fragmentation in short selling across multiple trading venues, aligning with the fragmentation of total exchange traded volume previously reported in the U.S. equity market (see, e.g., Chen and Duffie, 2021; Haslag and Ringgenberg, 2023; and O'Hara and Ye, 2011). This distribution of short volume across exchange groups highlights the importance of the comprehensive data collection and matching process to our analysis. The balanced distribution of liquidity-supplying short sales and liquidity-demanding short sales is consistent across each exchange group.⁹

Panel B of Table 1 shows correlations between different short selling activity measures and other stock and firm characteristics. We

⁸ The difference between the sum of the two components and the *ShortingFlow* is due to unsigned short sale transactions identified as neither liquidity-supplying nor liquidity-demanding (e.g., in cases of a crossed or locked market).

take the natural log of skewed variables, compute contemporaneous cross-sectional Pearson correlations each day, and report their time-series averages. Despite a positive correlation of 0.307 between *LSS* and *LDS*, a significant share of the cross-sectional variation remains unexplained. This suggests that other factors likely play a role in influencing the dynamics of the two types of short sales. Consistent with previous work (see, e.g., Boehmer and Wu, 2013 and Diether et al., 2009), we observe that short selling activity is on average contrarian: Higher contemporaneous and past week's returns coincide with higher short selling activity. Such contrarian trading has been at least partly attributed to short sellers' role as liquidity providers (Comerton-Forde et al., 2016). Our decomposition of shorting flow into *LSS* and *LDS* confirms this interpretation. The positive correlation between contemporaneous returns and short selling activity is present only for liquidity-supplying short sales, whereas liquidity-demanding short sales coincide with lower returns. A similar pattern emerges when we consider order imbalance as a proxy for stock specific buying pressure. Liquidity-supplying short sales increase with buying pressure relative to liquidity-demanding short sales. Diether et al. (2009) discuss the role of short sellers as, so called, opportunistic risk bearers by trading in stocks and time periods with elevated uncertainty. In this case, we expect short selling activity to correlate positively with intraday volatility and spreads. While the aggregate shorting flow does not show correlations consistent with this view, we again observe striking differences between those liquidity-supplying and liquidity-demanding short sales. *LSS* correlates positively with intraday volatility (*IntraVol*) and realized spread (*RS*), whereas the correlations of *LDS* with these two variables has the opposite sign. Liquidity-supplying short sales seem to step in to provide additional risk bearing capacity for stocks of elevated uncertainty. Taken together, short sellers tend to supply rather than demand liquidity in cases of high uncertainty.

Except for market capitalization, *LSS* and *LDS* exhibit similar correlations with other stock characteristics in our sample. For instance, both liquidity-supplying and liquidity-demanding short volumes correlate positively with active utilization (*ActiveUtilization*) and shorting fees *ShortingFee*. This positive relationship aligns with the notion that utilization and shorting fees are equilibrium outcome variables that increase with heightened shorting demand. The correlations with market capitalization (*Size*) suggest that short sellers are more likely to demand rather than supply liquidity for bigger stocks.

3. Empirical results

There is ample evidence that short selling activity predicts future negative returns (see, e.g., Asquith et al., 2005; Boehmer et al., 2008; Boehmer and Wu, 2013; Diether et al., 2009; and Engelberg et al., 2012). The literature implicitly treats informed short sales as liquidity-demanding (see, e.g., Comerton-Forde et al., 2016). Intuitively, liquidity-demanding short sales are likely to arise from informed traders, potentially those with short-lived information. On the other hand, while liquidity-supplying short sales would arise from market makers providing liquidity, it is also possible that these short sales come from other informed traders with longer-lived information and no need for immediacy (Boehmer et al., 2020). The previous literature's aggregation of these two categories masks the potentially heterogeneous impact of these short sales on predictability of returns. Our

⁹ We find no significant difference in the intraday distribution of *LSS* and *LDS* using 30 min trading intervals from 9:30 am to 4:00 pm. Fig. D.1 shows that both liquidity-supplying and liquidity-demanding shorting, as a percentage of total daily short volume, follow a U-shape distribution over the course of the trading day, similar to the patterns observed by Hu et al. (2025) in the Cboe exchange. The short sale ratios relative to total volume in each 30 min interval are stable over the trading day and similar in magnitude to the average daily short sale ratios.

Table 1
Summary statistics.

Panel A: Descriptives						
	Mean	Std Dev	10th	Median	90th	N
<i>Short sales</i>						
<i>ShortingFlow</i>	0.455	0.119	0.302	0.455	0.610	1,939
<i>LSS</i>	0.214	0.071	0.128	0.210	0.306	1,939
<i>LDS</i>	0.225	0.072	0.135	0.224	0.314	1,939
<i>ActiveUtilization</i>	13.900	18.841	0.744	6.554	35.763	1,934
<i>ShortingFee</i>	0.742	2.512	0.354	0.361	0.599	1,934
<i>Liquidity</i>						
<i>ILQ</i>	0.006	0.089	0.000	0.001	0.008	1,939
<i>RS</i>	3.857	13.242	-3.018	1.645	12.113	1,914
<i>PI</i>	10.364	14.189	1.002	6.836	23.431	1,914
<i>IntraVol</i>	1.515	8.072	0.028	0.277	2.586	1,914
<i>OIB</i>	-0.009	0.135	-0.161	-0.008	0.141	1,915
<i>Other characteristics</i>						
<i>Beta</i>	1.175	0.640	0.446	1.096	1.957	1,938
<i>Size</i>	10.9	35.6	5.0	2.2	21.7	1,939
<i>BM</i>	0.515	0.385	0.131	0.433	0.951	1,940
<i>IdioVol</i>	1.487	1.003	0.654	1.203	2.631	1,940
<i>R_t</i>	0.028	2.089	-1.798	0.019	1.834	1,940
<i>R_{t-4:t-1}</i>	0.112	4.186	-3.771	0.073	3.956	1,939

Panel B: Correlations			
	<i>ShortingFlow</i>	<i>LSS</i>	<i>LDS</i>
<i>ShortingFlow</i>	1.000	0.802	0.803
<i>LSS</i>	0.802	1.000	0.307
<i>LDS</i>	0.803	0.307	1.000
<i>ActiveUtilization</i>	0.350	0.302	0.253
<i>ShortingFee</i>	0.113	0.090	0.089
<i>ILQ</i>	-0.112	-0.021	-0.166
<i>RS</i>	-0.067	0.005	-0.105
<i>PI</i>	-0.008	0.057	-0.057
<i>IntraVol</i>	-0.060	0.034	-0.117
<i>OIB</i>	-0.054	0.156	-0.239
<i>Beta</i>	0.063	0.054	0.041
<i>Size</i>	0.038	-0.032	0.104
<i>BM</i>	-0.041	-0.028	-0.047
<i>IdioVol</i>	0.110	0.108	0.066
<i>Ret_t</i>	0.026	0.088	-0.042
<i>Ret_{t-4:t-1}</i>	0.050	0.061	0.023

This table reports the time-series average of the cross-sectional mean, standard deviation, 10/90th percentile, median, and average number of firms (N) of each variable. Short volume data are obtained from NYSE, Nasdaq, and Cboe. *Shortingflow* is the ratio of daily short volume to daily total volume. *LSS* and *LDS* are the corresponding ratios using liquidity-supplying and liquidity-demanding, respectively, short volumes. The direction of short trades are inferred from the algorithm of Lee and Ready (1991). The remaining variables are defined in A. The sample contains common stocks listed on the NYSE, Amex, and Nasdaq with price greater than \$1 and market capitalization above the 10th NYSE size percentile at end of day $t - 1$. The sample period is 2014 through 2018.

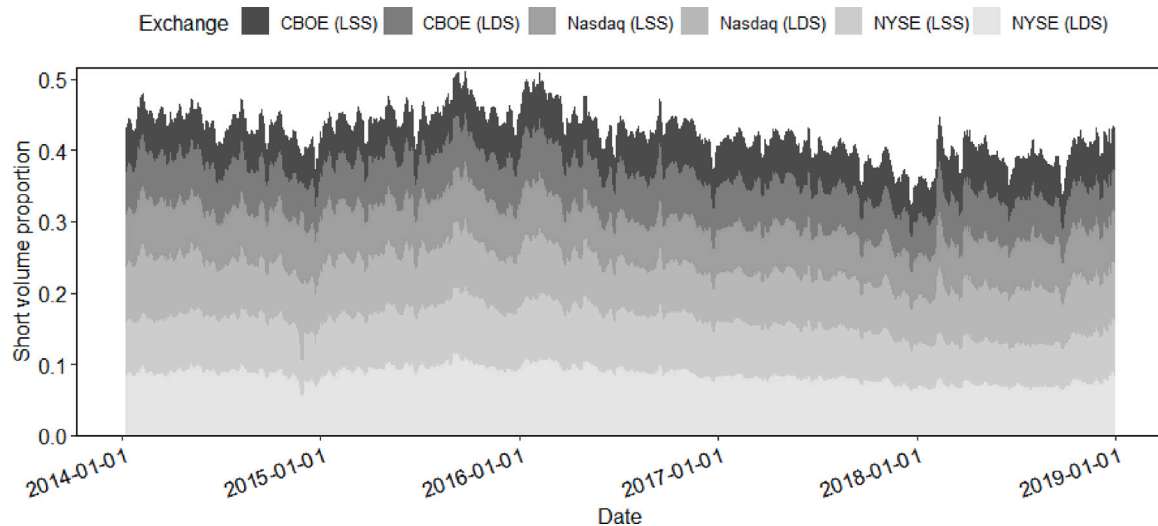


Fig. 1. Short volume across exchanges.

This figure shows the proportion of short volume relative to the total trading volume (five-day moving average) decomposed by exchange, and by liquidity-supplying (*LSS*) and liquidity-demanding (*LDS*) short volume ratios. Variables are defined in Appendix A. The sample period is 2014 through 2018.

classification of short sales into *LSS* and *LDS* allows us to analyze the potentially different predictability of returns from these two components. This analysis provides insights into which type of short selling is most informed and at which time horizon. By disaggregating the data, we can better understand the trading strategies of short sellers and how their informational advantage contributes to the efficient incorporation of information into stock prices.

3.1. Portfolio sorts

We start by examining the predictive ability of daily *LSS* and *LDS* by following the methodology of [Boehmer et al. \(2008\)](#). Specifically, we form quintile portfolios each day $t - 1$ based on that day's short volume ratio. We hold these portfolios for 21 days from day t through day $t + 20$. We use the overlapping portfolio approach of [Jegadeesh and Titman \(1993\)](#) to calculate overlapping period returns. The portfolios are value-weighted based on the market capitalization at the end of day $t - 1$. We calculate alphas from CAPM and a six-factor model that includes the five factors of [Fama and French \(2015\)](#) and the momentum factor.

We report cumulative six-factor alphas for spread portfolios formed based on sorts of *LSS* or *LDS* for future horizons varying from 1 to 21 days in [Fig. 2](#). Consistent with prior literature (see, e.g., [Boehmer et al., 2008](#) and [Engelberg et al., 2012](#)), we find that both *LSS* and *LDS* significantly and negatively predict next day t return, with daily six-factor alphas of around 0.10%. Increasing the holding period, we find that the predictive power of *LDS* weakens, with no statistically significant predictability after seven days. In contrast, we find that *LSS* is a strong negative predictor of future equity returns across all holding periods ranging from 1 to 21 days. For a 21-day holding period, the spread portfolio based on *LSS* has a cumulative six-factor alpha of almost 0.40%. Thus, this figure documents a striking empirical pattern: liquidity-supplying shorts negatively predict future equity returns, whereas liquidity-demanding shorts do not, at least for horizons of longer than a week.

We present returns and alphas on quintile portfolios and the spread portfolio for a horizon of 21 days in [Table 2](#). Panels A, B, and C sort on total *ShortingFlow*, *LSS*, and *LDS*, respectively. Panel A shows that heavily shorted stocks underperform lightly shorted stocks. This finding is robust whether we look at raw returns or risk-adjusted returns and is statistically significant with t -statistics ranging from -2.31 to -2.65 . Consistent with prior literature (see, e.g., [Boehmer et al., 2008](#)), we find that the performance difference between portfolios 1 and 5 is driven primarily by the outperformance of quintile 1 rather than by the underperformance of quintile 5.

Our analysis allows us to study the impact of these two kinds of short sales separately. Panel B for sorts on *LSS* shows that both returns and alphas decrease monotonically as one moves from quintile 1 to quintile 5. The spread portfolio has a return of -0.50% (t -statistic = -2.78) and a six-factor alpha of -0.38% (t -statistic = -3.31). The respective numbers in Panel C for sorts on *LDS* are -0.18% and -0.12% (both statistically insignificant). In fact, none of the quintile portfolios yields a statistically significant six-factor alpha in Panel C. Panel D shows the difference in returns to the spread portfolio for sorts on *LSS* and *LDS*. We find that the spread portfolio has economically and statistically significantly lower returns for sorts on *LSS* versus sorts on *LDS*. For example, the six-factor alpha difference is -0.26% (t -statistic = -2.68). These results suggest that the predictability of the short volume ratio, documented in the prior literature, is driven mostly by short trades that provide liquidity rather than demand it.

At first glance, our results might seem at odds with those of [Comerton-Forde et al. \(2016\)](#), who find that *LDS* trades are followed by lower returns relative to *LSS* trades. However, it is important to note that they conduct their analysis at intraday frequency, while our focus is on lower frequency returns. [Appendix C](#) provides more details on the reconciliation of our results with those of [Comerton-Forde et al.](#)

(2016). We find results broadly consistent with theirs, even over our extended sample and time period. The analysis in the appendix, thus, shows that any apparent discrepancy between our and their findings is due to the time horizon of analysis. Finally, it is also worth mentioning that [Fig. 2](#) shows that both *LSS* and *LDS* predict the next-day return negatively, which again supports the findings of [Comerton-Forde et al. \(2016\)](#) over the horizon that they study.

3.2. Cross-sectional regressions

While portfolio sorts are useful for gauging the economic importance of predictability, they do not allow us to control for potential confounding effects from other variables. Accordingly, we next employ [Fama and MacBeth \(1973\)](#) regressions to assess the predictive power of *LSS* and *LDS* in a multivariate setting. Following [Boehmer et al. \(2008\)](#) and [Engelberg et al. \(2012\)](#), on each day $t - 1$, we run the following cross-sectional regression:

$$R_{i,t:t+20} = \gamma_{0,t-1} + \gamma_{1,t-1} LSS_{i,t-1} + \gamma_{2,t-1} LDS_{i,t-1} + \gamma'_{3,t-1} X_{i,t-1} + \varepsilon_{i,t:t+20}, \quad (1)$$

where $R_{i,t:t+20}$ is the cumulative return from day t through day $t + 20$ and $X_{i,t-1}$ represents control variables measured with the latest available data as of day $t - 1$. The control variables include order imbalance, realized spread, price impact, intraday volatility, [Amihud \(2002\)](#) illiquidity, active utilization, shorting fees, market beta, log size, book-to-market ratio, idiosyncratic volatility, and past returns. To conform to the value-weighted portfolio sorts from the previous section, the regressions are value-weighted using market capitalization of day $t - 1$. We use [Newey and West \(1987\)](#) adjusted standard errors with 21 lags to adjust for serial correlation in the time series of the slope coefficients. In some specifications, we use panel regressions with firm and time fixed effects, with value-weighted observations and clustered standard errors. All independent variables are standardized each day by subtracting the cross-sectional mean and dividing by the cross-sectional standard deviation.

[Table 3](#) reports the results. Consistent with portfolio level analyses, we find that *LSS* has statistically significant negative predictive power for the 21-day-ahead cumulative return, while this is not the case for *LDS*. The slope coefficient on *LSS* is -0.18 (t -statistic = -3.27). As described previously, the independent variables in regressions are cross-sectionally standardized. An increase in *LSS* from the 10th to the 90th percentile, equivalent to a 2.56 standard deviation increase in a standard normal distribution, predicts a return difference of $-0.18 \times 2.56 = -0.46\%$ over the next 21 days. This finding aligns closely with our portfolio sort results in [Table 2](#).¹⁰ In contrast to *LSS*, the slope coefficient of *LDS* at -0.08 is economically small and statistically insignificant. When we include both *LSS* and *LDS* in specification (3), we find that *LSS* remains a significant negative predictor for future returns, whereas the *LDS* coefficient remains insignificant and even turns slightly positive.

Our results remain robust after the inclusion of several control variables, as is shown in specification (4) (we omit the coefficients on control variables for brevity). In panel regressions, following [Engelberg et al. \(2012\)](#), with time fixed effects (column 5) and time-plus-firm fixed effects (column 6), we find that the coefficient on *LSS* remains largely unchanged while the coefficient on *LDS* is positive. The regressions in column (6) that include stock fixed effects show that our result is due not merely to cross-sectional differences across stocks but arises also when *LSS* changes over time for a stock. Notably, the coefficient estimates on *LSS* are virtually identical in all specifications that we explore. In summary, the stock-level regressions lend further support to

¹⁰ We choose the 10th and 90th percentiles of the *LSS* distribution as they correspond to the medians of the top and bottom quintile portfolios from [Section 3.1](#).

Table 2
Returns to portfolios sorted on shorting flow.

	Panel A: <i>ShortingFlow</i>			Panel B: <i>LSS</i>			Panel C: <i>LDS</i>		
	Raw	CAPM	FF6	Raw	CAPM	FF6	Raw	CAPM	FF6
1	0.99 (1.99)	0.27 (3.33)	0.22 (3.08)	1.00 (2.04)	0.29 (3.75)	0.25 (3.55)	0.85 (1.71)	0.13 (1.57)	0.13 (1.56)
2	0.85 (1.65)	0.10 (1.83)	0.06 (1.27)	0.88 (1.71)	0.13 (2.35)	0.08 (1.95)	0.85 (1.67)	0.11 (2.01)	0.07 (1.46)
3	0.79 (1.53)	0.03 (0.79)	0.03 (0.74)	0.76 (1.46)	0.00 (0.01)	−0.01 (−0.14)	0.82 (1.59)	0.07 (1.71)	0.06 (1.50)
4	0.71 (1.35)	−0.05 (−0.71)	0.01 (0.15)	0.67 (1.28)	−0.09 (−1.23)	−0.03 (−0.54)	0.78 (1.48)	0.01 (0.21)	0.05 (1.06)
5	0.52 (0.95)	−0.25 (−1.76)	−0.10 (−1.16)	0.50 (0.91)	−0.28 (−2.09)	−0.13 (−1.67)	0.66 (1.22)	−0.12 (−0.98)	0.01 (0.09)
5−1	−0.46 (−2.31)	−0.53 (−2.65)	−0.32 (−2.48)	−0.50 (−2.78)	−0.57 (−3.23)	−0.38 (−3.31)	−0.18 (−1.09)	−0.25 (−1.51)	−0.12 (−0.88)
Panel D: <i>LSS − LDS</i>									
5−1				−0.32 (−2.97)	−0.32 (−3.00)	−0.26 (−2.68)			

At the end of each day $t-1$, we sort stocks into five portfolios based on their *ShortingFlow*, *LSS*, or *LDS*, and hold these portfolios from day t through day $t+20$. We then calculate overlapping period returns of these portfolios and report the performance numbers in this table. The columns labeled Raw, CAPM, and FF6 report the average raw returns, CAPM alphas, and Fama–French–Carhart six-factor alphas, respectively. We also report the performance of long-short 5–1 portfolios. The sample contains common stocks listed on the NYSE, Amex, and Nasdaq with price greater than \$1 and market capitalization above the 10th NYSE size percentile at the end of day $t-1$. Newey and West (1987) robust t -statistics are reported in parentheses. The sample period is 2014 through 2018.

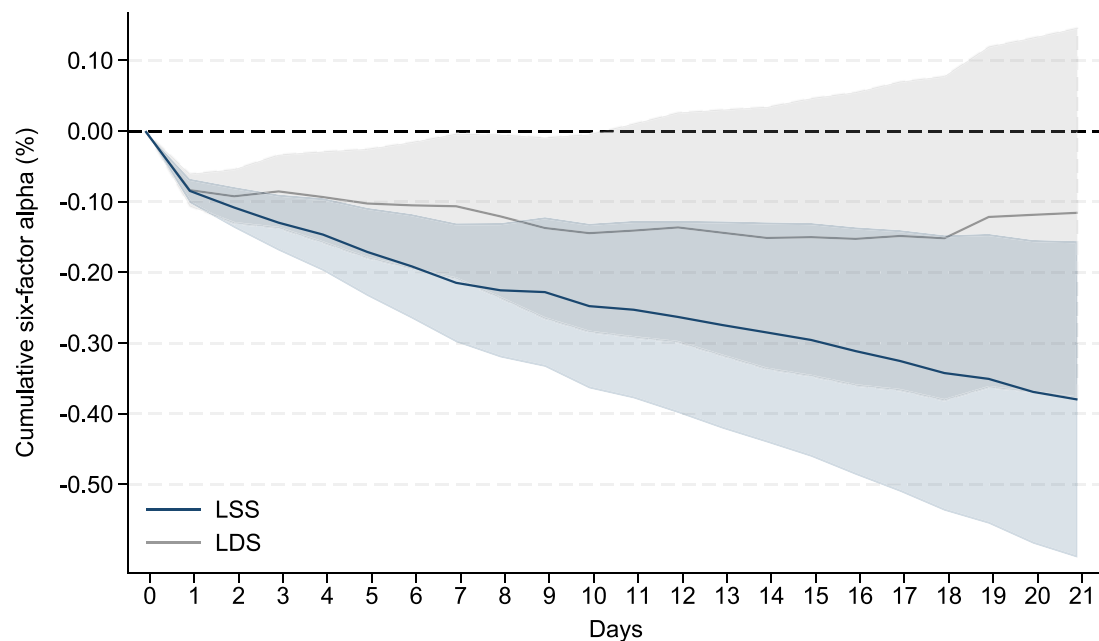


Fig. 2. Cumulative six-factor alpha.

This figure shows the cumulative six-factor alpha, expressed in percentage terms, for long-short portfolios based on liquidity-supplying (*LSS*) and liquidity-demanding (*LDS*) short volume ratios. Variables are defined in Appendix A. The shaded regions represent the 95% confidence intervals for the risk-adjusted return estimates. The sample period is 2014 through 2018.

the fact that the negative predictability from short sales is due primarily to liquidity-supplying short sales.

Overall, while we do find that liquidity-demanding short sales are associated with short-lived price impacts (as in Comerton-Forde et al., 2016), we show that liquidity-supplying short sales also predict negative returns, and do so more strongly and persistently. This finding challenges traditional market microstructure models, which associate informed trading with liquidity-demanding orders (e.g., Easley and O'Hara, 1987; Glosten, 1994; and Glosten and Milgrom, 1985).

In contrast, our findings align with a more recent, growing literature in which informed traders may optimally choose to provide liquidity by submitting limit orders. Models such as Kaniel and Liu

(2006) and Riccò et al. (2024) are particularly relevant in our context. Both frameworks allow informed traders to submit limit orders when their private information is long-lived, and show that such liquidity-supplying trades can be not only informative, but in some cases more informative than liquidity-demanding trades. Our evidence that *LSS* predict returns more strongly and over longer horizons is consistent with these predictions.¹¹

¹¹ Several other dynamic limit order book models (see, e.g., Brolley and Malinova, 2021; Goettler et al., 2009; and Roşu, 2020) also allow informed traders to provide liquidity. While these models differ in their assumptions about signal strength and execution risk, they broadly support the notion that

Table 3
Regressions of future returns on short volume.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>LSS</i>	−0.18 (−3.27)	—	−0.19 (−4.08)	−0.19 (−4.29)	−0.18 (−9.51)	−0.18 (−3.95)
<i>LDS</i>	—	−0.08 (−1.46)	0.02 (0.39)	0.01 (0.41)	0.07 (3.29)	0.09 (2.27)
Controls	No	No	No	Yes	Yes	Yes
Method	FMB	FMB	FMB	FMB	OLS	OLS
Time FE	—	—	—	—	Yes	Yes
Stock FE	—	—	—	—	No	Yes
<i>R</i> ²	0.57	0.61	1.06	13.12	22.42	25.76

We run regressions where the dependent variable is the 21-day-ahead cumulative return (from day t through day $t + 20$) and the independent variables (measured at day $t - 1$) are *LSS*, *LDS*, and other control variables. The control variables are *ActiveUtilization*, *ShortingFee*, *ILQ*, *RS*, *PI*, *IntraVol*, *OIB*, *Beta*, $\log(\text{Size})$, *BM*, *IdioVol*, R_{t-1} , and $R_{t-5:t-2}$. All variables are defined in A. All independent variables are standardized each day by subtracting the cross-sectional mean and dividing by the cross-sectional standard deviation. The sample contains common stocks listed on the NYSE, Amex, and Nasdaq with price greater than \$1 and market capitalization above the 10th NYSE size percentile at the end of day t . Columns (1)–(4) estimates are based on value-weighted Fama and MacBeth (1973) cross-sectional regressions, where we value-weight based on *Size* of day $t - 1$. We compute t -statistics (in parentheses) using Newey and West (1987) standard errors with 21 lags. Column (5) estimates are based on OLS regressions with day fixed effects and t -statistics are computed with standard errors clustered by day. Column (6) includes time and stock fixed effects and t -statistics are computed with standard errors clustered by day and stock. We omit the intercept and coefficients of control variables. The sample includes 2,385,741 observations from 2014 through 2018.

3.3. Cross-sectional differences in predictability

In an effort to identify whether our results are being driven by previously known results in the literature, we explore whether the return predictability from *LSS* is concentrated in stocks with certain characteristics. On day $t - 1$, we compute tercile ranks for each of the following conditioning variables: size, illiquidity, one-day return, one-week return, realized spread, price impact, intraday volatility, and order imbalance.¹² Subsequently, for each day $t - 1$, we run the following cross-sectional regression:

$$\begin{aligned}
 R_{i,t:t+20} = & \gamma_{0,t-1} + \gamma_{1,t-1} LSS_{i,t-1} + \gamma_{1,mid,t-1} LSS_{i,t-1} \times Mid_{i,t-1} \\
 & + \gamma_{1,high,t-1} LSS_{i,t-1} \times High_{i,t-1} + \gamma_{2,t-1} LDS_{i,t-1} \\
 & + \gamma_{2,mid,t-1} LDS_{i,t-1} \times Mid_{i,t-1} + \gamma_{2,high,t-1} LDS_{i,t-1} \times High_{i,t-1} \\
 & + Mid_{i,t-1} + High_{i,t-1} + \gamma'_{3,t-1} X_{i,t-1} + \epsilon_{i,t:t+20}, \quad (2)
 \end{aligned}$$

where $Mid_{i,t-1}$ and $High_{i,t-1}$ denote the mid and high tercile indicator variables on day $t - 1$ for stock i given a conditioning characteristic. The coefficients $\gamma_{1,t-1}$ and $\gamma_{2,t-1}$ correspond to *LSS* and *LDS*, respectively, for stocks in the low tercile. The coefficients on the tercile interactions can be interpreted as the differences in predictability between stocks in low and mid/high terciles. All regressions include the same control variables as those in column (4) of Table 3.¹³

The results presented in Table 4 show that no interaction term for any stock characteristic is statistically significant at the 5% level. This implies that predictability from *LSS* does not differ for stocks with different characteristics. For ease of interpretation, we provide the sum of the coefficients and their interactions with *High* in the last two rows of this table. These coefficients show the predictability for the subsample of stocks in the *High* tercile. We find that across all characteristics, predictability from *LSS* is negative and statistically significant, while this is not the case for *LDS*.¹⁴

informed investors do not always trade aggressively and may strategically post limit orders under certain conditions.

¹² In unreported results, we also test whether liquidity-supplying short sales are particularly informative about future returns in the case of larger retail buying pressure, which may be perceived as likely uninformed. Our findings indicate that the predictive power of *LSS* does not vary with changes in retail order imbalance.

¹³ When including the tercile indicators based on the conditioning stock characteristic, we omit the corresponding continuous version of that variable.

¹⁴ One exception is the interaction with *Size*. We find that, for this stock characteristic, the coefficient on *LDS* is negative and statistically significant,

3.4. Robustness checks

We examine the robustness of results from Section 3.2 by using various sample splits in Panel A of Table 5. The two columns under the heading “Univariate” show the regression coefficients of *LSS* and *LDS* when the variables are used as single predictors in two separate regressions (as in columns (1) and (2) of Table 3). The two columns below the heading “Bivariate” show the regression coefficients of *LSS* and *LDS* when used jointly as predictors in one regression (as shown in column (3) of Table 3). The columns under the heading “With controls” show the regression coefficients of *LSS* and *LDS* when we augment the regression with additional control variables (as shown in column (4) of Table 3).

Different rows of Panel A of Table 5 show results for different samples. Row (1) shows the baseline regression results from Table 3. Row (2) uses the sample period from 2014 through June 2016. Row (3) uses the sample period from July 2016 through 2018. Row (4) removes stocks below \$5 or below the 20th NYSE size percentile. Row (5) uses all stocks (without price or micro-cap filter). Row (6) covers a sample of the largest 500 stocks based on market capitalization. Rows (7) through (9) consider different horizons of future returns as dependent variables. Row (10) uses the five-day average of *LSS* and *LDS* at day t as the main independent variable.

We find that, with hardly any exceptions, the coefficient on *LSS* is negative and statistically significant while that on *LDS* is statistically insignificantly different from zero.¹⁵ The results are similar across two subsamples (rows (2) and (3)), for different filters on stocks (rows (4) through (6)) and for different horizons of future returns (rows (7) through (9)). Row (9) in particular shows that the predictability extends to 40 days. Thus our results are not driven by specific time periods, specific samples of stocks, or particular calculations of variables.

To further examine whether the predictive power of liquidity-supplying trades is unique to short sellers, we extend our analysis to include the corresponding measures for long sales (Comerton-Forde et al., 2016). Using *LSS*, *LDS*, and daily volume of all buyer- and seller-initiated trades from WRDS IID, we infer two additional variables: the liquidity-supplying long sales ratio (*LSSL*) and the

while the coefficient on $LDS \times High$ is positive and statistically significant. This suggests that small stocks do show negative predictability of returns following *LDS* while large stocks do not.

¹⁵ The coefficient on *LSS* is statistically insignificant in row (3) while the coefficient on *LDS* is statistically significant in row (2) both in univariate regressions.

Table 4
Return predictability across different stock characteristics.

	<i>Size</i>	<i>ILQ</i>	R_t	$R_{t-5:t-1}$	<i>RS</i>	<i>PI</i>	<i>IntraVol</i>	<i>OIB</i>
<i>LSS</i>	−0.13 (−3.65)	−0.22 (−3.90)	−0.21 (−3.77)	−0.19 (−3.25)	−0.19 (−3.92)	−0.20 (−3.53)	−0.21 (−3.59)	−0.20 (−3.34)
<i>LSS</i> × <i>Mid</i>	0.07 (1.85)	0.13 (1.95)	−0.00 (−0.06)	0.03 (0.70)	−0.03 (−1.04)	0.05 (1.06)	0.05 (0.90)	−0.06 (−1.50)
<i>LSS</i> × <i>High</i>	−0.08 (−1.38)	0.08 (1.32)	0.08 (1.78)	−0.00 (−0.01)	0.08 (1.81)	0.05 (0.83)	0.08 (0.98)	0.04 (0.71)
<i>LDS</i>	−0.13 (−3.11)	0.04 (0.72)	0.00 (0.08)	−0.02 (−0.39)	−0.01 (−0.15)	0.06 (0.86)	0.06 (0.98)	0.03 (0.51)
<i>LDS</i> × <i>Mid</i>	−0.01 (−0.40)	−0.18 (−2.80)	0.02 (0.47)	0.04 (0.75)	0.09 (2.31)	−0.13 (−1.79)	−0.20 (−2.99)	0.02 (0.46)
<i>LDS</i> × <i>High</i>	0.18 (2.46)	−0.11 (−1.57)	0.00 (0.02)	0.05 (1.00)	−0.06 (−1.07)	−0.14 (−1.66)	−0.13 (−1.53)	0.03 (0.46)
<i>LSS</i> + <i>LSS</i> × <i>High</i>	−0.22 (−4.10)	−0.13 (−4.44)	−0.13 (−2.70)	−0.19 (−3.88)	−0.11 (−2.71)	−0.15 (−4.57)	−0.14 (−3.51)	−0.16 (−3.56)
<i>LDS</i> + <i>LDS</i> × <i>High</i>	0.06 (0.91)	−0.07 (−1.88)	0.00 (0.10)	0.03 (0.62)	−0.07 (−1.55)	−0.09 (−1.75)	−0.07 (−1.43)	0.05 (1.20)

We run Fama and MacBeth (1973) cross-sectional regressions where the dependent variable is the 21-day-ahead cumulative return (from day t through day $t + 20$) and the independent variables (measured at day $t - 1$) are *LSS*, *LDS*, and other control variables as in column (4) of Table 3, except that we include interactions of *LSS* and *LDS* with tercile indicator variables based on stock characteristics (shown in each column). All variables are defined in A. Except for indicator variables, all independent variables are standardized each day by subtracting the cross-sectional mean and dividing by the cross-sectional standard deviation. The sample contains common stocks listed on the NYSE, Amex, and Nasdaq with price greater than \$1 and market capitalization above the 10th NYSE size percentile at the end of day $t - 1$. We compute t -statistics (in parentheses) using Newey and West (1987) standard errors with 21 lags. We omit the intercept and coefficients on control variables. The last two rows present the sum of the coefficients on *LSS* or *LDS* and their interaction with the High tercile indicator variable. The sample includes 2,385,741 observations from 2014 through 2018.

liquidity-demanding long sales ratio (*LDLS*).¹⁶ This decomposition of long sales volume allows us to address two key questions: first, and most importantly, whether *LSS* remains a strong predictor of returns after controlling for the two long sales ratios and, second, whether the decomposition of long sales exhibits patterns similar to those observed for short sales.

In Panel B of Table 5, we include two additional specifications that control for *LSSL* and *LDLS*, respectively, alongside *LSS* and *LDS*.¹⁷ Consistent with our main findings, *LSS* remains both statistically and economically significant in negatively predicting future returns. In numbers not reported in the table, we also find that the coefficient on *LSSL* is 0.08 (t -statistic = 1.83) and the coefficient on *LDLS* is −0.05 (t -statistic = −1.45) suggesting that these trades are on average uninformed. These findings highlight the unique role of liquidity-supplying short sellers in incorporating information into prices, a role not shared by their counterparts in long sales.

Overall, the results of this section emphasize the robust predictability of future returns associated with liquidity-supplying short sales, contrasted by the absence of such predictability for liquidity-demanding short sales and the two long sales counterparts.

4. What explains the return predictability?

The evidence reported in Section 3 indicates that liquidity-supplying short sales negatively predict long-term equity returns. Moreover, we do not find such predictive power for liquidity-demanding short sales. In this section we discuss and test three hypotheses to explain our results.

¹⁶ *LSSL* is defined as the ratio of buyer-initiated long sales to total trading volume, while *LDLS* is the ratio of seller-initiated long sales to total trading volume.

¹⁷ The sum of the four components (*LSS*, *LDS*, *LSSL*, and *LDLS*) is by definition equal to one. Therefore, due to multicollinearity concerns, *LSSL* and *LDLS* are added separately in two different specifications. Similarly, we omit the order imbalance (*OIMB*) variable from these specifications.

The first hypothesis builds on previous work and the general perception that short sellers are informed (see, e.g., Asquith et al., 2005; Boehmer et al., 2008; Christophe et al., 2004; Christophe et al., 2010; Desai et al., 2006; Diether et al., 2009; Figlewski, 1981; and Senchack and Starks, 1993). Some informed traders may find it optimal under certain conditions to submit limit orders and supply liquidity (see, e.g., Chakravarty and Holden, 1995; Garriott and Riordan, 2024; O'Hara, 2015; and Seppi and Kumar, 1994). They may prefer limit orders to trade based on fundamental news, especially if these are long-lived (Collin-Dufresne and Fos, 2015; Kaniel and Liu, 2006; and Riccò et al., 2024). As short sellers have become more profitable trading on long-term information that is gradually incorporated into prices (Wang et al., 2020), the return predictability based on *LSS* is consistent with informed trading.

In addition to the informed trading hypothesis, two alternative mechanisms could also explain the reported return predictability: market making and opportunistic risk-bearing. Diether et al. (2009) show that short selling activity correlates positively with contemporaneous returns. Comerton-Forde et al. (2016) find that this contrarian trading at daily horizon is driven by buyer-initiated short sales. These findings are consistent with short sellers stepping in to serve as market makers. When stock prices rise following a sequence of buy orders, market makers reduce their long positions and may even shift to short positions to continue providing liquidity. This surge in selling activity caters to buyers willing to pay for immediacy. As buying pressure subsides, stock prices are likely to revert and compensate short sellers for their market-making activity.

The third hypothesis to explain the results is that short sellers are opportunistic risk-bearers (Diether et al., 2009). For stocks and time periods with higher price uncertainty, short sellers increase their activity to opportunistically supply liquidity. Once the heightened uncertainty is resolved, the stock price is expected to revert to its long-term level, thus providing a return to the short seller for bearing this risk. In the following subsections, we run a number of empirical exercises to shed light on which of the three non-mutually exclusive hypotheses most likely applies to liquidity-supplying short sales.

Table 5
Return predictability across different samples.

		Univariate		Bivariate		With controls	
		<i>LSS</i>	<i>LDS</i>	<i>LSS</i>	<i>LDS</i>	<i>LSS</i>	<i>LDS</i>
Panel A: Changes to baseline of Table 3							
(1)	Baseline	−0.18 (−3.27)	−0.08 (−1.46)	−0.19 (−4.08)	0.02 (0.39)	−0.19 (−4.29)	0.01 (0.41)
(2)	First-half	−0.22 (−3.32)	−0.14 (−2.78)	−0.20 (−2.86)	−0.04 (−0.84)	−0.15 (−1.98)	−0.04 (−0.86)
(3)	Second-half	−0.15 (−1.61)	−0.02 (−0.21)	−0.19 (−2.92)	0.08 (1.05)	−0.23 (−5.10)	0.07 (1.29)
(4)	No <\$5 or 20% micro	−0.17 (−3.18)	−0.07 (−1.39)	−0.18 (−4.06)	0.02 (0.42)	−0.18 (−4.23)	0.01 (0.40)
(5)	No exclusions	−0.24 (−3.25)	−0.11 (−1.34)	−0.25 (−4.04)	0.03 (0.44)	−0.25 (−4.43)	0.03 (0.50)
(6)	Largest 500	−0.15 (−2.92)	−0.04 (−0.77)	−0.18 (−3.96)	0.06 (1.16)	−0.19 (−3.99)	0.04 (1.06)
(7)	$R_{t:t+4}$	−0.05 (−3.08)	−0.01 (−0.76)	−0.06 (−3.86)	0.01 (0.87)	−0.05 (−3.61)	0.02 (1.17)
(8)	$R_{t+5:t+20}$	−0.09 (−2.13)	−0.03 (−0.64)	−0.11 (−3.02)	0.03 (0.80)	−0.11 (−3.25)	0.03 (1.02)
(9)	$R_{t:t+39}$	−0.26 (−3.41)	−0.13 (−1.86)	−0.27 (−3.97)	0.01 (0.16)	−0.24 (−4.38)	0.00 (0.08)
(10)	5-day average of X	−0.19 (−1.77)	−0.07 (−0.65)	−0.25 (−2.83)	0.09 (1.01)	−0.21 (−3.00)	0.05 (0.88)
Panel B: With additional controls							
(1)	With <i>LSSL</i>	–	–	–	–	−0.12 (−3.38)	0.02 (0.50)
(2)	With <i>LDLS</i>	–	–	–	–	−0.18 (−4.23)	−0.03 (−1.04)

We run Fama and MacBeth (1973) cross-sectional regressions where the dependent variable is the 21-day-ahead cumulative return (from day t through day $t + 20$) and the independent variables (measured at day $t - 1$) are *LSS*, *LDS*, and other control variables as in Table 3. The two columns under the heading “Univariate” show the regression coefficients of *LSS* and *LDS* when the variables are used as single predictors in two separate regressions (as in columns (1) and (2) of Table 3). The two columns below the heading “Bivariate” show the regression coefficients of *LSS* and *LDS* when used jointly as predictors in one regression (as in column (3) of Table 3). The columns below “With controls” show the regression coefficients of *LSS* and *LDS* when we augment the regression with additional control variables (as in column (4) of Table 3). All variables are defined in A. All independent variables are standardized each day by subtracting the cross-sectional mean and dividing by the cross-sectional standard deviation. Different rows use different samples. Row (1) shows the baseline regression results from Table 3. The baseline sample contains common stocks listed on the NYSE, Amex, and Nasdaq with price greater than \$1 and market capitalization above the 10th NYSE size percentile at the end of day $t - 1$ and the sample period is from 2014 through 2018. Row (2) uses the sample period from 2014 through June 2016. Row (3) uses the sample period from July 2016 through 2018. Row (4) removes stocks below \$5 or below the 20th NYSE size percentile. Row (5) uses all stocks (without price or micro-cap filter). Row (6) uses a sample of the largest 500 stocks based on market capitalization. Rows (7) through (9) consider different horizons of future returns as dependent variable. Row (10) uses the five-day average of *LSS* and *LDS* at day $t - 1$ as main dependent variables. In Panel B, we add additional control variable, *LSSL* or *LDLS* (please see text for description) to the list of control variables. We compute t -statistics (in parentheses) using Newey and West (1987) standard errors with 21 lags.

4.1. The role of market making and opportunistic risk bearing

To test whether the return predictability based on *LSS* can be at least partly attributed to short sellers’ compensation for market making activity and/or opportunistic risk bearing, we apply an approach similar to that of Boehmer et al. (2021b) to our hypotheses. In the first stage, we decompose *LSS* into three components, using cross-sectional regressions for each day $t - 1$:

$$LSS_{i,t-1} = \underbrace{\beta_{t-1} R_{i,t-1} + \gamma_{t-1} OIB_{i,t-1}}_{\text{voluntary market making (mm)}} + \underbrace{\delta_{t-1} IntraVol_{i,t-1}}_{\text{opportunistic risk bearing (rb)}} + \underbrace{\alpha_{t-1} + \varepsilon_{i,t-1}}_{\text{residual (res)}}. \quad (3)$$

For each day $t - 1$, we estimate the exposure of *LSS* to daily return, *Ret*, order imbalance, *OIB*, and intraday volatility, *IntraVol*. Then, we calculate the following three components of *LSS* for each day $t - 1$, using the cross-sectional regression coefficients of day $t - 1$ from the first stage:

$$\begin{aligned} LSS_{mm,i,t-1} &= \beta_{t-1} R_{i,t-1} + \gamma_{t-1} OIB_{i,t-1} \\ LSS_{rb,i,t-1} &= \delta_{t-1} IntraVol_{i,t-1} \\ LSS_{res,i,t-1} &= \alpha_{t-1} + \varepsilon_{i,t-1}. \end{aligned} \quad (4)$$

Motivated by the model and the empirical evidence of Comerton-Forde et al. (2016), we label the variation of *LSS* that relates to a stock’s return and order imbalance as the market making component of *LSS* (*LSS_{mm}*). Similar to Diether et al. (2009), we denote the variation of liquidity-supplying short selling associated with a stock’s intraday volatility as the risk bearing component of *LSS* (*LSS_{rb}*). The remaining variation, *LSS_{res}*, potentially contains other relevant information about future returns reflected in liquidity-supplying short sales.

We report the results of the decomposition from Eq. (3) in Panel A of Table 6. Coefficients on all three explanatory variables are positive and highly economically and statistically significant at all conventional levels. These results are consistent with our expectations and the cross-correlations reported in Table 1. Liquidity-supplying short selling is contrarian and increases with stronger buying pressure. Moreover, *LSS* also increases with heightening stock-level uncertainty as proxied by intraday volatility. In sum, *LSS* contains variation that can be explained by both market making activity and opportunistic risk bearing.

In the second stage, we employ Fama and MacBeth (1973) regressions of Eq. (1) to predict the 21-day-ahead stock returns using the three components of *LSS*. As in Section 3.2, we use Newey and West (1987) adjusted standard errors with 21 lags to adjust for serial correlation in the time series of the slope coefficients and weight observations by

Table 6
Decomposing LSS volume.

Panel A: First stage regression				
	Ret	OIB	$IntraVol$	R^2
Slope	33.58 (22.34)	17.69 (92.72)	0.09 (6.76)	9.72
Panel B: Future return predictability				
	(1)	(2)	(3)	(4)
LSS_{mm}	-0.04 (-1.37)	–	-0.04 (-1.37)	1.04 (0.46)
LSS_{rb}	–	-0.06 (-1.14)	-0.06 (-1.22)	-0.01 (-0.26)
LSS_{res}	-0.18 (-3.35)	-0.18 (-3.35)	-0.18 (-3.35)	-0.17 (-4.36)
Controls	No	No	No	Yes
R^2	1.07	0.70	1.19	11.33

In Panel A, we regress LSS_{t-1} on contemporaneous return (Ret_{t-1}), order imbalance (OIB_{t-1}), and intraday volatility ($IntraVol_{t-1}$). We then use the fitted values of this regression to define market-making component (LSS_{mm}), risk-bearing component (LSS_{rb}), and denote the residual as LSS_{res} . Panel B reports descriptive statistics on these components. Panel C uses these components in Fama and MacBeth (1973) cross-sectional regressions where the dependent variable is the 21-day-ahead cumulative return (from day t through day $t+20$) as in Table 3. The control variables in column (4) in Panel C are the same as those in column (5) of Table 3 except that we do not include the variables used in the first-stage regression, namely Ret , OIB , and $IntraVol$. All variables are defined in A. All independent variables are standardized each day by subtracting the cross-sectional mean and dividing by the cross-sectional standard deviation. The sample contains common stocks listed on the NYSE, Amex, and Nasdaq with price greater than \$1 and market capitalization above the 10th NYSE size percentile at the end of day $t-1$. We compute t -statistics (in parentheses) using Newey and West (1987) standard errors with 21 lags. The sample includes 2,385,741 observations from 2014 through 2018.

the stock's market capitalization at day $t-1$. We include the three components of LSS in different specifications and report the estimates in Panel B of Table 6. All independent variables are cross-sectionally standardized each day by subtracting the cross-sectional and dividing by cross-sectional standard deviation.

In column (1), we find that the coefficient on LSS_{res} is negative at -0.18 (t -statistic = -3.75). There is remarkable similarity in the coefficient on LSS_{res} in Table 6 and the coefficient on LSS in Table 3. Column (1) shows that the coefficient on LSS_{mm} is -0.04 (t -statistic = -1.37). Hence, we cannot reject the null hypothesis that the market making component of LSS does not contribute significantly to the predictive power of liquidity-supplying short sales. In column (2), we use the opportunistic risk-bearing component of LSS to predict cross-sectional stock returns and find similar results: the coefficient on LSS_{rb} is -0.06 (t -statistic = -1.14). Results change only slightly when we add all three components in column (3). In the last specification, next to the three components, we also add our remaining control variables. The residual component, LSS_{res} , remains the only economically and statistically significant component of LSS when predicting future stock returns.

In sum, we find that market making activity and opportunistic risk bearing of liquidity-supplying short sales explain a significant part of the cross-sectional variation in LSS . However, neither of these two components contains significant predictive power for stock returns. The predictability of returns is driven entirely by the residual component, consistent with the hypothesis that liquidity-supplying short sales contain material information about future stock prices.

4.2. Liquidity provision and news

In this section, we further explore the predictive power of LSS in the cross-section of stock returns. In particular, we ask whether the return predictability shown in Section 3 arises from the informational advantage and skill of liquidity-supplying short sellers in identifying mispriced stocks. To address this question, we employ a framework akin to that of Engelberg et al. (2012). The rationale behind our test is as follows: if the predictive ability of liquidity-supplying short sales is indeed due to an informational advantage, characterized by trading in overpriced stocks and avoiding underpriced ones, then this mispricing

should be corrected upon the release of news. In other words, LSS should predict returns particularly on days with news releases.¹⁸

To test this hypothesis, we compile a sample of news events using the RavenPack Analytics (RPA) database and study the short selling patterns around public news releases. The RPA database contains a comprehensive sample of firm specific news stories from thousands of news sources, such as Dow Jones Newswires, The Wall Street Journal, Barron's, and other leading financial websites. To identify important news events specific to a given firm, we require both the *Relevance* and *Event Relevance* measure to be above 70 (out of 100).¹⁹ Moreover, we consider only news events with an *Event Similarity Day* score above 90 (up to 365 days), thereby requiring at least 90 days since a similar event was detected in the past. Following Reed et al. (2020), we remove (1) news stories relating to trading and market activity, such as "technical-analysis", "stock-prices", and "insider-trading"; and (2) news in the categories "revenues" and "investor-relations", so as to avoid a large overlap with earnings news announcements days. Furthermore, we augment the news sample with earnings announcement dates from the Institutional Brokers' Estimate System (IBES) if they are not already included in the RPA database. We define a stock-day news indicator variable ($News$) as one if a news event occurred on that day for a particular firm and zero otherwise. Following Engelberg et al. (2012), if a news story is released after 4:00 pm, then we define the following trading day as the event day.

We then extend the Fama and MacBeth (1973) regressions of Section 3.2 by including the news indicator variable and interaction terms with our two short selling measures, LSS and LDS . We report the results in Table 7. As a starting point, column (1) simply replicates the bivariate regression of 21-day-ahead returns on LSS and LDS from column (3) of Table 3. Column (2) shows that, while the coefficient on the $News$ indicator variable is small and insignificant, the coefficient on the interaction term, $LSS \times News$, is relatively large (-0.17), and

¹⁸ Boehmer et al. (2020) and Jank et al. (2021) use a similar setting to better understand the informational content of short sales and how it varies around news events and earnings announcement days.

¹⁹ *Event Relevance* provides information on the prominence of an event within a news story, whereas *Relevance* focuses on how strongly a company relates to the underlying news story. Our empirical results are robust to restricting both relevance scores to above 90 or even to 100.

Table 7
Relation between future returns, short volume, and news.

	(1)	(2)	(3)	(4)
<i>LSS</i>	−0.19 (−3.91)	−0.12 (−2.81)	−0.12 (−2.81)	−0.07 (−2.20)
<i>LDS</i>	0.02 (0.38)	0.00 (0.04)	0.00 (0.04)	−0.00 (−0.11)
<i>LSS</i> × <i>News</i>	–	−0.17 (−2.64)	–	–
<i>LDS</i> × <i>News</i>	–	0.09 (1.09)	–	–
<i>News</i>	–	0.08 (0.81)	–	–
<i>LSS</i> × <i>PosNews</i>	–	–	0.14 (1.86)	0.09 (1.33)
<i>LSS</i> × <i>NeutNews</i>	–	–	−0.17 (−2.29)	−0.16 (−2.50)
<i>LSS</i> × <i>NegNews</i>	–	–	−0.27 (−3.71)	−0.24 (−3.10)
<i>LDS</i> × <i>PosNews</i>	–	–	0.01 (0.14)	0.02 (0.21)
<i>LDS</i> × <i>NeutNews</i>	–	–	0.14 (1.57)	0.14 (2.00)
<i>LDS</i> × <i>NegNews</i>	–	–	0.09 (1.33)	0.12 (1.71)
Controls	No	No	No	Yes
<i>R</i> ²	1.04	2.37	6.27	16.61

We run Fama and MacBeth (1973) cross-sectional regressions where the dependent variable is the 21-day-ahead cumulative return (from day t through day $t + 20$) and the independent variables (measured at day $t - 1$) are *LSS*, *LDS*, and other control variables as in Table 3, except that we include interactions of *LSS* and *LDS* with news indicator variables. *News* is an indicator variable equal to one if a firm-specific news event occurs on day t , and zero otherwise. *PosNews*, *NeutNews*, and *NegNews* are indicator variables equal to one for firm-specific news event days, if the announcement day t return falls into the top, middle three, and bottom quintile of daily returns, respectively; and zero otherwise. All other variables are defined in A. Except for indicator variables, all independent variables are standardized each day by subtracting the cross-sectional mean and dividing by the cross-sectional standard deviation. The sample contains common stocks listed on the NYSE, Amex, and Nasdaq with price greater than \$1 and market capitalization above the 10th NYSE size percentile at the end of day $t - 1$. We compute t -statistics (in parentheses) using Newey and West (1987) standard errors with 21 lags. We omit the intercept and coefficients on control variables. The sample includes 2,324,138 observations from 2014 through 2018.

statistically significant with a t -statistic of -2.64 . In the same specification, the coefficient on *LSS* by itself is -0.12 (t -statistic = -2.81). Thus, the predictive power of *LSS* more than doubles on news days. In contrast, the coefficient on the *LDS* × *News* is 0.09 and not statistically different from zero. These results suggest that the return predictability of *LSS* stems from the informational advantage of liquidity-supplying short sellers, with their predictive power being most pronounced on days when information is released and embedded into stock prices.

Next, to study whether the increased predictability of *LSS* depends on news content, we decompose the *News* indicator variable into three distinct components: positive news, neutral news, and negative news. Specifically, following the approach of Engelberg et al. (2012), we define *PosNews* and *NegNews* as indicator variables that take the value of one when news occurs and the firm's return falls within the highest and lowest quintiles, respectively, of all event-returns for that day and zero otherwise. *NeutNews* is assigned a value of one when news occurs but the firm's return is neither in the top nor the bottom quintile and zero otherwise.²⁰

The last two columns of Table 7 show the results of regressions in which we interact *LSS* and *LDS* with these news sentiment indicator variables. Column (3), for instance, suggests that the predictive ability seen before for *LSS* is driven most strongly by negative news events (coefficient on *LSS* × *NegNews* is -0.27 with a t -statistic of -3.71), and, to a lesser extent, by neutral news events (coefficient on *LSS* ×

NeutNews is -0.17 with a t -statistic of -2.29). The magnitude of these coefficients suggests that the *LSS* predictability more than triples on days with negative news events. The coefficients are similar in column (4), where we include additional control variables. We also find in Table 7 that the coefficient on *LSS* × *PosNews* is positive, unlike the coefficients on interactions with *NeutNews* and *NegNews*, but statistically indistinguishable from zero (t -statistic of 1.33 in column (4)). It is also worth noting that the predictive power of *LDS*, as opposed to *LSS*, does not differ across news content types, and is not statistically significant for any type of news content.

At a broad level, the analysis in this section shows that liquidity-supplying short sales are particularly informative about future returns on news days relative to non-news days. This is also consistent with Engelberg et al. (2012), who find that short sales (the sum of liquidity-supplying and liquidity-demanding short sales in their setting) have more predictive power for returns on news days. At a more detailed level, this section shows that liquidity-supplying short sales are most informative in the presence of negative or neutral news relative to positive news.

4.3. News and short sellers' motivation

Section 4.1 shows that the residual component of liquidity-supplying shorts is the most informative about future returns. Section 4.2 shows that the predictive content of liquidity-supplying shorts is highest around news days, and especially so around negative news days. In this section, we combine the liquidity-supplying short sale decomposition, i.e., market-making (*LSS_{mm}*), risk-bearing (*LSS_{rb}*), and residual (*LSS_{res}*) components from Table 6 with the news decomposition from

²⁰ This approach, as highlighted by Engelberg et al. (2012) and Reed et al. (2020), offers an advantage over sentiment scores based on positive and negative word counts because it incorporates information relative to the market's prior expectations.

Table 8
Relation between future returns, components of *LSS* volume, and news.

	(1)	(2)	(3)	(4)
LSS_{mm}	-0.03 (-1.12)	-0.05 (-1.93)	-0.05 (-1.93)	2.49 (1.52)
LSS_{rb}	-0.07 (-1.32)	-0.06 (-1.40)	-0.06 (-1.40)	-0.02 (-0.40)
LSS_{res}	-0.19 (-3.94)	-0.11 (-2.43)	-0.11 (-2.43)	-0.08 (-2.64)
$LSS_{mm} \times News$	-	0.06 (1.04)	-	-
$LSS_{rb} \times News$	-	0.39 (0.84)	-	-
$LSS_{res} \times News$	-	-0.23 (-3.02)	-	-
<i>News</i>	-	0.04 (0.37)	-	-
$LSS_{mm} \times PosNews$	-	-	0.10 (1.28)	0.12 (1.60)
$LSS_{mm} \times NeutNews$	-	-	0.13 (2.05)	0.11 (2.09)
$LSS_{mm} \times NegNews$	-	-	-0.08 (-0.94)	-0.03 (-0.40)
$LSS_{rb} \times PosNews$	-	-	2.43 (2.54)	2.15 (2.81)
$LSS_{rb} \times NeutNews$	-	-	0.70 (1.04)	0.51 (1.28)
$LSS_{rb} \times NegNews$	-	-	0.28 (0.27)	0.15 (0.18)
$LSS_{res} \times PosNews$	-	-	0.07 (0.76)	0.03 (0.35)
$LSS_{res} \times NeutNews$	-	-	-0.25 (-2.98)	-0.24 (-3.34)
$LSS_{res} \times NegNews$	-	-	-0.25 (-3.32)	-0.24 (-3.20)
Controls	No	No	No	Yes
R^2	1.61	3.32	7.76	16.39

We run [Fama and MacBeth \(1973\)](#) cross-sectional regressions where the dependent variable is the 21-day-ahead cumulative return (from day t through day $t + 20$) and the independent variables (measured at day $t - 1$) are components of *LSS* from [Table 6](#), news indicator variables from [Table 7](#), and other control variables from [Table 6](#). All other variables are defined in [A](#). Except for indicator variables, all independent variables are standardized each day by subtracting the cross-sectional mean and dividing by the cross-sectional standard deviation. The sample contains common stocks listed on the NYSE, Amex, and Nasdaq with price greater than \$1 and market capitalization above the 10th NYSE size percentile at the end of day $t - 1$. We compute t -statistics (in parentheses) using [Newey and West \(1987\)](#) standard errors with 21 lags. We omit the intercept and coefficients on control variables. The sample includes 2,324,138 observations from 2014 through 2018.

Table 7. The goal is to evaluate how short sellers' motivation might shed light on the source of their informational advantage

Table 8 reports the results. As a starting point, column (1) simply replicates column (3) of [Table 6](#), where we see that the residual component drives the vast majority of *LSS*'s predictability. In column (2), we add the interaction terms, and we see that the residual component of *LSS* drives the vast majority of predictability on news days. In particular, the coefficient on $LSS_{res} \times News$ is -0.23 , which is strongly statistically significant (t -statistic = -3.02), unlike all of the other news interactions. In terms of magnitude, comparing the coefficient of -0.11 on the standalone version of LSS_{res} to the effect of $LSS_{res} \times News$ shows that future returns are roughly three times as large on news days than on non-news days. In other words, as with previous results, short sellers' information advantage while supplying liquidity accrues mostly on news days, and the residual component of their liquidity provision seems to be the key driver of that advantage. Together, these findings further support informed trading as the most plausible explanation for the return predictability associated with *LSS*.

Columns (3) and (4) show a finer-grained decomposition in which news is split into positive, neutral, and negative news. Among numerous interaction variables, the baseline effect of LSS_{res} is far stronger than those of LSS_{mm} and LSS_{rb} . When looking at the interaction terms

involving LSS_{res} , negative and neutral news are more negative than those with positive news. To demonstrate the magnitude of this effect, the coefficient estimate in column (3) for $LSS_{res} \times NegNews$ is -0.25 (t -statistic = -3.32). The magnitude of the coefficient on $LSS_{res} \times NeutNews$ is similar, as is the statistical significance, but the magnitude and significance of the coefficient on $LSS_{res} \times PosNews$ are much lower.

In general, we see very little statistical significance for coefficients on either the market making or risk bearing component of liquidity-supplying short sales. Thus, the residual component of liquidity-supplying short sales drives most of the results, providing further evidence that the return predictability result is likely due to informed trading by liquidity-supplying short sellers.

On the other hand, we notice that $LSS_{rb} \times PosNews$ has a statistically significant coefficient estimate in both columns (3) and (4). Since the LSS_{rb} component has no effect on future returns in the models without interactions in columns (1) and (2), we suspect that the positive effect in the last two columns comes from a stronger influence of outliers, due to the finer disaggregation and resulting reliance on a smaller number of observations for each category in models (3) and (4).

We next explore how the volume of liquidity-supplying short sales varies around news days. Our empirical approach follows [Engelberg](#)

et al., 2012). Specifically, we run a series of panel regressions with stock and day fixed effects, where the dependent variable is one of the components of LSS . The independent variable takes the value of one if a news story occurs and zero otherwise. Across the specifications, we vary the timing of the dependent variable relative to the news event to examine short-volume changes around news. For example, -1 indicates the change in LSS observed one day prior to the news event. Lastly, we follow Engelberg et al. (2012) and include two lags of daily returns as control variables in all regression specifications. We plot the coefficient estimates on the news indicator variables from these regressions as bars in Fig. 3. In Panel A we limit the sample to negative news announcements, and in Panel B we limit the sample to positive news announcements.

We first focus on the residual component of liquidity-supplying short sales as the dependent variable. Panel A shows a statistically significant increase of LSS_{res} a few days before the release of negative news, peaking on the day of the event. This evidence indicates that liquidity-supplying short sellers exhibit some anticipation of negative news events, although the economic magnitude is relatively small compared to the cross-sectional standard deviation of LSS . When we consider positive news separately in Panel B, we find evidence that the ratio of liquidity-supplying short sales to other trades is small in the days leading up to a news event. While we would expect that informed short sellers open fewer short positions ahead of (or on the day of) positive news releases, interestingly, we find that the residual component of LSS increases on the day of the positive event. This finding is consistent with our results in Tables 7 and 8 showing that LSS_{res} exhibits the weakest return predictability on positive news days compared to the remaining news days in our sample.

Next, we study the dynamics of the market-making component of LSS around news days. For both negative and positive news events, this component remains quite stable before and after news days. However, there is a striking effect on the day of the announcement of news. Negative news days show significantly less market-making activity by short sellers, whereas we observe the opposite for positive news days. This finding is consistent with LSS_{mm} capturing contrarian trading by short sellers. On days with positive news, short sellers step in and sell when buying pressure is high. Conversely, in the case of negative news, when there is overall more selling than buying pressure, we observe significantly less informal market making by short sellers.²¹

Overall, our results support the interpretation that the return predictability based on liquidity-supplying short sales reflects the informational advantage held by short sellers over other market participants. The return predictability is particularly pronounced (1) on days when fundamental information is disclosed to the market and (2) when we exclude the market-making and risk-bearing components from our short selling measure.

4.4. Short sellers and cross-sectional return anomalies

Our results so far indicate that the predictive power of short sales for the cross-section of equity returns stems from informed liquidity provision by short sellers. Moreover, the graphical representation of the return predictability for varying holding periods in Fig. 2 suggests that these short sales trade on information that is slowly incorporated into stock prices.

In more recent and dynamic market microstructure theories (see, e.g. Kaniel and Liu, 2006 and Riccò et al., 2024) informed investors may strategically supply liquidity by submitting limit orders, particularly when their information pertains to long-horizon mispricing and

when execution risk is low. In these settings, patient informed traders optimize execution by waiting for liquidity takers to arrive while avoiding the immediate costs of demanding liquidity.

To further shed light on the nature of the information that liquidity-supplying short sellers exploit, we study whether their short sales align with well-known cross-sectional equity anomalies. These anomaly variables, derived from accounting, price, and/or trading data, are known to predict returns over extended periods — often spanning months to years — and are thought to reflect slow-moving information about firm fundamentals (see, e.g. Arnott et al., 2023 and Ehsani and Linnainmaa, 2022). If liquidity-supplying short sellers are indeed informed investors who trade based on long-term mispricing signals, we would expect them to disproportionately sell stocks in the short legs of these anomalies and avoid stocks in the long legs.

To this end, we examine 10 prominent anomaly variables that the previous literature has found to be related to the cross-section of stock returns.²² If liquidity-supplying short sellers are informed investors who understand the return predictability based on these anomalies, then we expect these short sellers to sell stocks in the short leg of these anomalies and to avoid stocks in the long leg.

We follow the procedure of Wang et al. (2020) in testing this hypothesis. Each anomaly variable is signed, based on prior literature, such that the future mean returns increase in the signed characteristic. We sort all stocks in our sample into deciles based on each signed anomaly variable. We follow previous literature in forming the rebalancing and holding frequencies. For anomalies constructed using annual Compustat data, we form portfolios at the end of each June in year y by using accounting data from the fiscal year ending in calendar year $y - 1$. Furthermore, we hold the portfolio from July of year y through June of year $y + 1$. For anomalies constructed using quarterly Compustat data (such as those using ROE), we form portfolios at the end of quarter q using data from the fiscal quarter ending in calendar quarter $q - 1$ and hold the portfolio over calendar quarter $q + 1$. Lastly, for anomalies constructed using monthly CRSP data, we form portfolios every month m and hold the portfolio for the next month $m + 1$. Using this portfolio formation procedure, for each stock-month and for each anomaly, we calculate the decile rank for that stock based on its corresponding portfolio assignment. Next, for each day t , we sort stocks into quintiles based on either LSS or LDS . For each shorting quintile and for each anomaly, we subsequently compute the average anomaly decile rank at the end of the previous month. Since decile 10 (1) of each anomaly represents the long (short) leg, we expect the average rank of the high shorting quintile to be significantly lower than the average rank of the low shorting quintile.

We show the results in Panel A of Table 9. We find that the average anomaly rank of LSS quintile 1 is higher than that of LSS quintile 5 for all anomalies. For instance, for the anomaly based on Net Stock Issues, the average rank of quintiles 1 and 5 of LSS portfolios is 7.07 and 6.11, respectively. This means that stocks for which liquidity-supplying short sales are high (quintile 5) are more likely to correspond to stocks that have low Net Stock Issues and thus are overpriced. The difference in these ranks is -0.96 with a t -statistic of -11.67 calculated using Newey and West (1987) errors with 60 lags. In fact, the difference in average anomaly ranks between LSS quintiles 5 and 1 is negative and statistically significant across all anomalies. The average difference across all anomalies is -0.72 (t -statistic = -23.65).

Panel A also shows that the difference in average anomaly ranks between LDS quintiles 5 and 1 is negative and statistically significant for 9 out of 10 anomalies. In the last column of Panel A, we compute the difference-in-differences between LSS and LDS and find that the

²¹ We also study the opportunistic risk-bearing component of liquidity-supplying short sales, LSS_{rb} , around both negative and positive news days. There is no significant variation in LSS_{rb} across day -5 through day 5 surrounding the event dates. For the sake of clarity, we have omitted these results from Fig. 3. However, they are available in Fig. D.2.

²² We use the following anomalies: Net Stock Issues, Composite Equity Issues, Accruals, Net Operating Assets, Asset Growth, Investments-to-Assets, Distress, Momentum, Gross Profitability, Return on Assets. Variable definitions are provided in Appendix A.

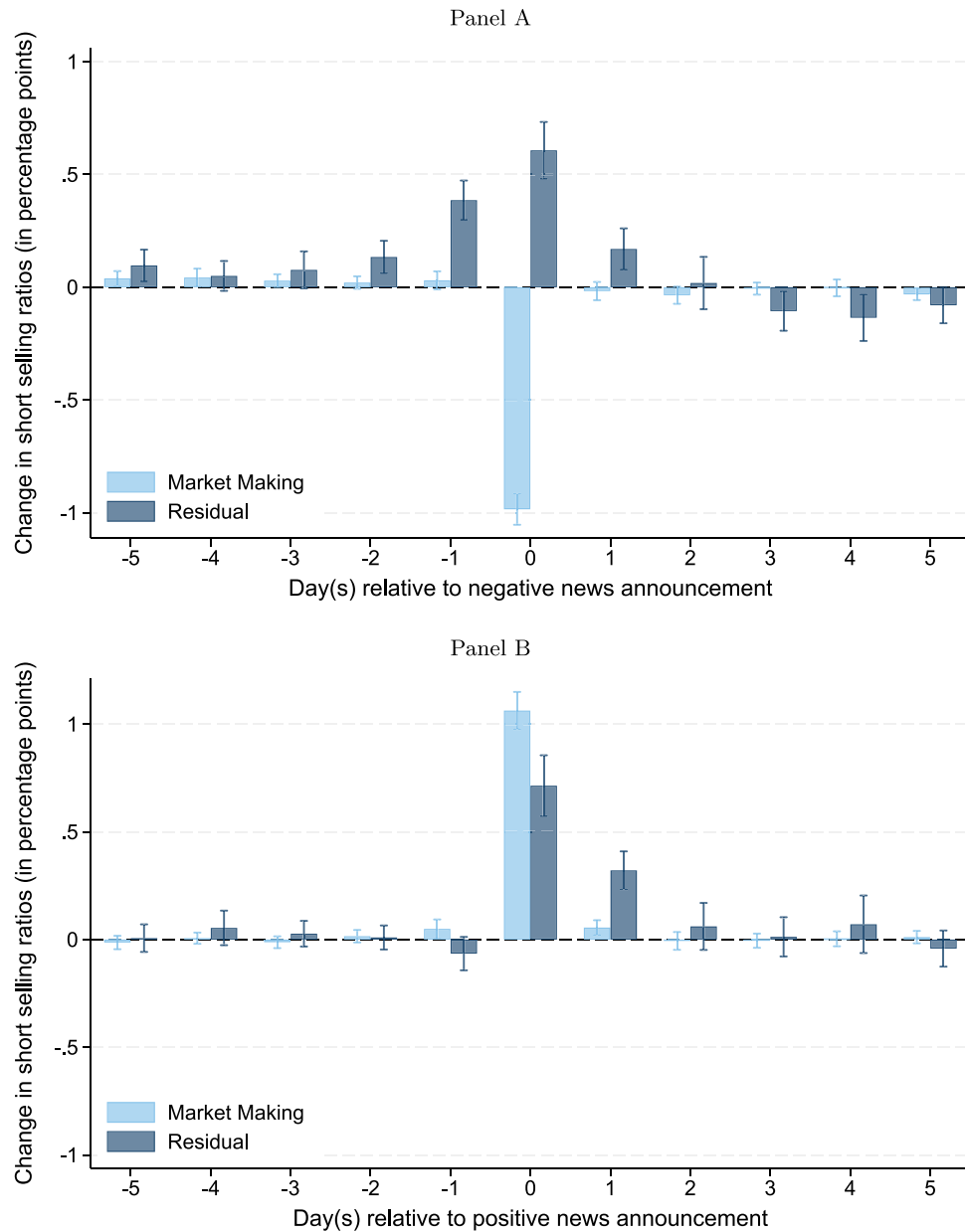


Fig. 3. Components of LSS around news event days.

This figure shows the change of the residual and the liquidity-supplying component of LSS in percentage points around firm-specific news events. Each bar is based on a panel regression with stock and day fixed effects. The dependent variable is one of the daily components of LSS and the independent variable of interest is an indicator variable that takes the value one if a news story occurs and zero otherwise. In each regression we vary the timing of the dependent variable relative to the news event, to examine short-volume changes around news. For example, $t - 1$ indicates that the dependent variable is observed one day prior to the news event. To control for the response of short sellers to past returns, we include five-day lags of daily returns (where the lags are relative to the timing of the dependent variable). In Panel A and Panel B we examine changes in components of LSS around negative and positive news events, respectively. We define a news event as negative (positive) if the announcement day return is in the bottom (top) quintile of returns on a given day, respectively. The sample period is 2014 through 2018.

spread in average anomaly rank for LSS is larger than for LDS . Eight of the ten anomalies exhibit statistically significant differences at the five percent level.

It may appear odd that we find significant differences in average anomaly rank even for LDS quintiles, given the earlier absence of evidence regarding long-term predictability from LDS . Note, however, that LSS and LDS are cross-sectionally correlated with a correlation of 0.307 (see Table 1). To decipher the pure effect of LSS and LDS , we make one modification to the above procedure. Specifically, we orthogonalize LDS each day t by cross-sectionally regressing $LDS_{i,t}$ on

$LSS_{i,t}$ and taking the residuals. We call these residuals $LDS^\perp$; by construction they are purged of the effect of LSS . We construct a similar orthogonalized version of LSS by cross-sectionally regressing $LSS_{i,t}$ on $LDS_{i,t}$ and calling the residuals $LSS^\perp$. Panel B of Table 9 shows the results using $LSS^\perp$ and $LDS^\perp$ as sorting variables. Comparing Panels A and B, we find similar results for LSS and its orthogonalized counterpart $LSS^\perp$. The results are, however, markedly different for LDS and $LDS^\perp$. Only one anomaly shows a negative spread between quintiles 1 and 5 of $LDS^\perp$ that is also statistically significant. The

Table 9
Anomalies and short volume.

	1	5	5-1		1	5	5-1		5-1	
Panel A: Using unorthogonalized LSS and LDS										
	LSS				LDS				$LSS - LDS$	
Net Stock Issues	7.07	6.11	-0.96	(-11.67)	6.71	6.42	-0.29	(-2.69)	-0.67	(-8.83)
Comp. Equity Issues	6.90	5.99	-0.90	(-10.37)	6.54	6.17	-0.37	(-3.91)	-0.53	(-12.50)
Accruals	5.72	5.62	-0.10	(-1.94)	5.73	5.69	-0.04	(-0.95)	-0.05	(-1.44)
Net Operating Assets	5.77	5.49	-0.27	(-3.51)	5.78	5.68	-0.11	(-1.48)	-0.16	(-3.49)
Asset Growth	5.75	5.40	-0.35	(-4.83)	5.69	5.55	-0.15	(-2.29)	-0.20	(-3.02)
Investments-to-Assets	5.91	5.23	-0.68	(-11.90)	5.93	5.30	-0.63	(-10.72)	-0.05	(-0.69)
Distress	8.33	6.66	-1.67	(-24.38)	8.06	6.97	-1.09	(-9.60)	-0.58	(-7.76)
Momentum	6.15	5.40	-0.75	(-8.17)	6.02	5.53	-0.49	(-5.84)	-0.26	(-5.36)
Gross Profitability	5.86	5.37	-0.49	(-9.05)	5.73	5.40	-0.34	(-4.47)	-0.15	(-3.52)
Return on Assets	7.04	5.96	-1.08	(-12.09)	6.82	6.12	-0.69	(-6.44)	-0.39	(-6.84)
Average	6.45	5.72	-0.72	(-23.65)	6.30	5.88	-0.42	(-9.44)	-0.31	(-10.66)
Panel B: Using orthogonalized $LSS^\perp$ and $LDS^\perp$										
	$LSS^\perp$				$LDS^\perp$				$LSS^\perp - LDS^\perp$	
Net Stock Issues	6.99	6.09	-0.90	(-14.65)	6.32	6.61	0.29	(3.78)	-1.19	(-10.71)
Comp. Equity Issues	6.86	6.06	-0.80	(-15.51)	6.25	6.41	0.15	(2.73)	-0.95	(-17.07)
Accruals	5.68	5.56	-0.11	(-2.37)	5.63	5.69	0.06	(1.50)	-0.17	(-2.42)
Net Operating Assets	5.77	5.55	-0.22	(-3.62)	5.76	5.77	0.01	(0.22)	-0.23	(-2.73)
Asset Growth	5.73	5.40	-0.33	(-5.38)	5.60	5.60	-0.00	(-0.01)	-0.33	(-3.16)
Investments-to-Assets	5.79	5.35	-0.44	(-7.35)	5.78	5.43	-0.35	(-6.28)	-0.09	(-0.88)
Distress	8.07	6.77	-1.30	(-22.88)	7.47	7.31	-0.15	(-1.61)	-1.15	(-9.22)
Momentum	6.10	5.53	-0.57	(-8.27)	5.88	5.76	-0.12	(-1.93)	-0.45	(-5.30)
Gross Profitability	5.80	5.45	-0.35	(-9.11)	5.61	5.50	-0.10	(-1.62)	-0.24	(-3.52)
Return on Assets	6.90	6.03	-0.87	(-15.34)	6.41	6.38	-0.03	(-0.40)	-0.84	(-10.93)
Average	6.37	5.78	-0.59	(-26.32)	6.07	6.05	-0.02	(-0.73)	-0.57	(-12.69)

Each month within the sample period, we assign decile ranks to stocks based on an anomaly. All variables are defined in A. Decile 1 (10) is the group of stocks associated with lower (higher) future returns. On each trading day of the subsequent month, we independently sort stocks into quintiles based on LSS and LDS . This table presents the average rank of quintiles 1, 5, and the difference between them. We report Newey and West (1987) adjusted (using 60 lags) t -statistics for the differences in parentheses. Panel A uses unorthogonalized LSS and LDS while Panel B uses orthogonalized $LSS^\perp$ and $LDS^\perp$. $LSS^\perp$ is the residual from cross-sectional regression each day of LSS on LDS . $LDS^\perp$ is the residual from cross-sectional regression each day of LDS on LSS . The sample period is from 2014 through 2018.

average difference across all anomalies for $LDS^\perp$ is only -0.02 (t -statistic = -0.73). We conclude that liquidity-supplying shorting flows are reactive to information embedded in anomalies, in the sense that investors short more of stocks deemed as overvalued than stocks deemed as undervalued. The same is not case for liquidity-demanding shorting flows.

Lastly, we decompose LSS into three components as before: market making, risk bearing, and the residual. We repeat our analysis by sorting on these components and computing the average anomaly ranks. We show the results in Table 10. For LSS_{res} , we find that the average decile anomaly rank for the high shorting quintile is significantly lower than that for the low shorting quintile for all anomalies. For LSS_{mm} , we find significant differences for 7 out of 10 anomalies. Lastly, for LSS_{rb} , we find significant differences for 8 out of 10 anomalies. Across the 10 anomalies, the average 5–1 difference in anomaly ranks is -0.71 , -0.16 , and -0.45 for LSS_{res} , LSS_{mm} , and LSS_{rb} , respectively. As in the previous sections, we find that the residual component of short selling supply is the one most closely aligned with well-known profitability of anomaly strategies.

4.5. Short sellers and venue choice

So far, our analysis aggregates short sales across trading venues into daily stock-level measures of liquidity-supplying (LSS) and liquidity-demanding (LDS) short volume. This aggregation obscures the fact that trading is spread across multiple trading venues, and these trading venues are heterogeneous in ways that could be related to the patterns of liquidity and informativeness we observe in the results above.²³ In

particular, our previous results suggest that liquidity-supplying short sellers trade patiently on long-lived information and submit limit orders (see, e.g., Kaniel and Liu, 2006 and Riccò et al., 2024). However, not all limit orders are equally passive: depending on the venue, traders may submit limit orders more aggressively to expedite execution. Therefore, to more directly examine whether liquidity-supplying shorts are truly patient, as our findings imply, we exploit differences across trading venues, where distinct fee structures create incentives for patient versus aggressive liquidity provision.

One key difference between trading venues is their fee model, which directly affects trading behavior and liquidity provision. The eleven exchange venues that we cover in our sample have mainly operated under two primary fee structures during our sample period: the maker-taker fee model and the inverted (taker-marker) fee model. In maker-taker venues, liquidity providers who submit limit orders receive a rebate, thereby reducing the effective cost of liquidity provision, while liquidity takers incur a fee. In contrast, inverted venues impose a fee on liquidity providers and pay a rebate to liquidity takers.²⁴ These structural differences create important strategic considerations for informed traders. In particular, while submitting limit orders is generally associated with liquidity provision, limit orders on inverted venues tend to be more aggressive because traders pay a fee and thus often seek faster execution (Garvey and Qin, 2024).

Thus, the venue choice provides a natural way to distinguish patient informed trading from faster, more aggressive trading. Maker-taker venues, by subsidizing liquidity supply and lowering the effective cost of patiently posting limit orders (Malinova and Park, 2015), are particularly attractive to traders acting on slow-moving, long-term

²³ For example, Fig. 1 shows the proportion of short volume relative to the total trading volume decomposed by exchange, indicating considerable market fragmentation on U.S. equity exchanges.

²⁴ See, among others, Battalio et al. (2016), Comerton-Forde et al. (2019), Di Maggio et al. (2020), and Garvey and Qin (2024), for more discussion on maker-taker versus inverted venues.

Table 10
Anomalies and components of LSS volume.

	1	5	5-1		1	5	5-1		1	5	5-1	
	LSS_{mm}				LSS_{rb}				LSS_{res}			
Net Stock Issues	6.65	6.21	-0.43	(-13.44)	6.63	5.70	-0.93	(-5.11)	7.03	6.13	-0.89	(-10.83)
Comp. Equity Issues	6.43	6.39	-0.05	(-1.18)	6.33	5.77	-0.56	(-4.54)	6.89	5.99	-0.90	(-10.42)
Accruals	5.60	5.71	0.11	(4.04)	5.68	5.32	-0.36	(-4.36)	5.73	5.62	-0.11	(-2.30)
Net Operating Assets	5.74	5.63	-0.11	(-4.23)	6.15	5.97	-0.18	(-4.07)	5.76	5.49	-0.27	(-3.66)
Asset Growth	5.68	5.46	-0.22	(-8.23)	5.64	5.51	-0.12	(-2.73)	5.74	5.41	-0.33	(-4.88)
Investments-to-Assets	5.68	5.46	-0.22	(-8.40)	5.73	5.70	-0.03	(-1.12)	5.89	5.23	-0.66	(-12.46)
Distress	7.56	7.17	-0.39	(-7.89)	7.70	6.18	-1.52	(-5.07)	8.31	6.69	-1.62	(-24.28)
Momentum	5.85	5.84	-0.01	(-0.26)	5.87	5.78	-0.09	(-1.10)	6.15	5.38	-0.77	(-8.87)
Gross Profitability	5.65	5.57	-0.07	(-1.83)	5.62	5.54	-0.08	(-3.16)	5.84	5.36	-0.48	(-9.07)
Return on Assets	6.51	6.29	-0.22	(-4.42)	6.48	5.80	-0.68	(-5.00)	7.03	5.95	-1.07	(-11.82)
Average	6.13	5.97	-0.16	(-8.17)	6.18	5.73	-0.45	(-5.01)	6.43	5.72	-0.71	(-23.07)

Each month within the sample period, we rank stocks into ten deciles based on an anomaly. All variables are defined in A. Decile 1 (10) is the group of stocks associated with lower (higher) future returns. On each trading day in the subsequent month, we independently sort stocks into quintiles based on LSS_{mm} , LSS_{rb} , or LSS_{res} , from Table 6 in ascending order. This table presents the average rank (from 1 through 10) of each anomaly for quintiles 1, 5, and the difference between them. We report Newey–West adjusted (using 60 lags) t -statistics for the differences in parentheses. The sample period is 2014 through 2018.

Table 11
Regressions of future returns on short volume: Maker-taker versus taker-maker venues.

	(1)	(2)	(3)	(4)	(5)	(6)
LSS_{mt}	–0.21 (–3.57)	–0.19 (–3.66)	–	–	–0.21 (–2.87)	–0.20 (–3.12)
LDS_{mt}	–0.03 (–0.48)	–0.03 (–0.88)	–	–	–0.04 (–0.69)	–0.04 (–0.99)
LSS_{inv}	–	–	–0.11 (–2.54)	–0.11 (–3.01)	–0.03 (–0.46)	–0.03 (–0.79)
LDS_{inv}	–	–	–0.05 (–1.09)	–0.05 (–1.52)	0.05 (1.08)	0.05 (1.21)
Controls	No	Yes	No	Yes	No	Yes
R^2	1.28	13.28	1.01	13.13	2.17	13.83

We run regressions where the dependent variable is the 21-day-ahead cumulative return (from day t through day $t+20$) and the independent variables (measured at day $t-1$) are LSS_{mt} , LDS_{mt} , LSS_{inv} , LDS_{inv} , and other control variables. The control variables are *ActiveUtilization*, *ShortingFee*, *ILQ*, *RS*, *PI*, *IntraVol*, *OIB*, *Beta*, $\log(\text{Size})$, *BM*, *IdioVol*, R_{t-1} , and $R_{t-5:t-2}$. All variables are defined in A. All independent variables are standardized each day by subtracting the cross-sectional mean and dividing by the cross-sectional standard deviation. The sample contains common stocks listed on the NYSE, Amex, and Nasdaq with price greater than \$1 and market capitalization above the 10th NYSE size percentile at the end of day t . All estimates are based on value-weighted Fama and MacBeth (1973) cross-sectional regressions, where we value-weight based on *Size* of day $t-1$. We compute t -statistics (in parentheses) using Newey and West (1987) standard errors with 21 lags. We omit the intercept and coefficients of control variables. The sample includes 2,385,741 observations from 2014 through 2018.

information. In contrast, inverted venues are better suited for informed traders who use limit orders but prioritize execution speed over patience. In summary, while the optimal trading strategy may depend on the time horizon of the information, our framework suggests that maker-taker venues are particularly favorable to the stealth trading of longer-run information. Consequently, we expect the negative return predictability of liquidity-supplying short sales to be more pronounced in maker-taker venues than in inverted venues.

To test this idea, we construct separate LSS and LDS measures using trades from maker-taker (mt) and inverted (inv) venues, and assess their ability to predict future 21-day returns.²⁵ We show the results in Table 11. Consistent with our expectations, we find that LSS_{mt} has statistically significant negative predictive power for the 21-day-ahead cumulative return. In column (1), we find that the coefficient on LSS_{mt} is -0.21 (t -statistic = -3.57). In column (2), we add control variables, and find similar results. In column (3), we run a bivariate regression of 21-day-ahead returns on LSS and LDS of inverted trading venues. We find that the coefficient on LSS_{inv} is -0.11 (t -statistic = -2.54). Controlling for additional regressors in column (4) displays a similar picture. In column (5), we regress future 21-day returns jointly on

LSS and LDS for both types of venues. LSS_{mt} has a coefficient of -0.21 (t -statistic = -2.87), whereas all other short volume types are statistically insignificant. In particular, the coefficient on LSS_{inv} at -0.03 is much smaller than that on LSS_{mt} , suggesting that liquidity-supplying short sales on maker-taker venues are more informed than those on inverted venues. In column (6), we find similar results after accounting for control variables. Overall, we find that LSS on maker-taker venues is driving the return predictability, consistent with our thinking above. Because liquidity provision on maker-taker venues best reflects the strategic limit order submission envisioned in dynamic market microstructure models, our evidence that liquidity-supplying short sales on maker-taker venues predict future returns is consistent with these models of patient informed trading, in particular with those of Kaniel and Liu (2006) and Riccò et al. (2024).

5. Conclusion

Our research provides significant insights into the role of short sellers in financial markets, particularly in relation to their impact on market liquidity and their informativeness about trades. By distinguishing between liquidity-supplying and liquidity-demanding short sales, we challenge the conventional wisdom that only those demanding liquidity are informed. We show that liquidity-supplying short sellers are, in fact, better at predicting future stock returns, particularly over longer holding periods. Our study aligns with more recent theoretical work that posits a dual role for informed traders, one that includes liquidity provision as a strategy to capitalize on long-lived information.

²⁵ In our sample period, NYSE, NYSE American (until June 2017), NYSE Arca, Nasdaq, Nasdaq PSX, BZX, and EDGX use a maker-taker fee model, whereas NYSE National, Nasdaq BX, BYX, and EDGA (until May 2017) use an inverted fee model.

Regulators have long faced a difficult challenge in regulating short sellers. Regulators must balance the benefits of short-selling—such as enhanced liquidity and price discovery—with concerns about market stability and potential price distortions. While restrictions such as short-sale bans or price tests are often introduced to prevent excessive downward pressure, empirical evidence suggests that they can also reduce market efficiency and harm liquidity. These trade-offs, as discussed in [Edwards et al. \(2024\)](#), are likely to persist well into the future.

The fact that the same short sellers both supply liquidity and improve market efficiency adds significantly to the challenge. Our paper shows that short sellers who trade in the direction of their information, which presumably affects prices, are, more often than not, doing so through liquidity-providing trades. In other words, our results show that the liquidity provision aspect of short selling is not separable from the price discovery aspect of short selling, at least along the liquidity supply/demand dimension.

Regulations relying on microstructure price mechanisms, including price tests such as the uptick rule, assume informed short sellers are liquidity demanders. It is through the mechanism of identifying liquidity-demanding short sales that these regulations work to achieve their aim of reducing the price impact of short sales (see [Jain, Jain, McNish, and McKenzie, 2013](#)). Since the results in our paper indicate that liquidity-supplying short sales can be more informed than liquidity-demanding short sales, the price mechanisms central to these regulations might be ill-suited to reduce the price impact of short selling.

Given our results, future regulation might prioritize disclosure-based regulations over price tests. For example, new regulation might take the form of the US SEC's recent expansion of short-sale disclosure, which amends the consolidated audit trail (CAT) to include an additional short sale related field, or they might take the form of the UK's recent proposal to publicly disclose aggregated short positions.²⁶ These examples of disclosure-based regulations do not rest on the assumption that informed short sellers are always liquidity-demanding short sellers.

CRediT authorship contribution statement

Amit Goyal: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Adam V. Reed:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Esad Smajlbegovic:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Amar Soebhag:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

Amar Soebhag has a part-time position at Robeco Quantitative Investing. Robeco did not provide any data or funding for this research. Remaining authors have nothing to disclose.

Appendix A. Variable definitions

See [Table A.1](#).

²⁶ For the US proposal, see “[ReportingofSecuritiesLoans](#)”, and for the UK proposal, see “[GovernmentResponse](#).”

Appendix B. Matching short sale transactions

We match our intraday short sale data with all trades and quotes from the same period using the TAQ data. Specifically, as in [Barber et al. \(2024\)](#), we use the WRDS Consolidated Trades (WCT) files, which provides us with the National Best Bid and Offer (NBBO) at the time of each trade. The WCT files include the NBBO quotes following the convention established by [Holden and Jacobsen \(2014\)](#). We exclude trades associated with locked or crossed bid–ask spreads from the WCT files, also in line with [\(Holden and Jacobsen, 2014\)](#). For all remaining trades in the WCT files, we classify trades following the [\(Lee and Ready, 1991\)](#) (henceforth LR) method, using trade prices and the prevailing NBBO.

We next match each short sale trade across all U.S. exchanges to its corresponding trade in the WCT universe. Our matching procedure allows us to classify each short sale as either buyer- or seller-initiated based on the NBBO and the LR methodology. Our procedure consists of three steps, detailed below.

Step 1: Exact Matching

In the first step, we perform an exact match between short sales and WCT trades based on the following grouping variables: ticker symbol, exchange code, trade price, trade size, and date/time (at the second or tenth-of-a-second level, depending on the provided time precision). We match trades that have identical values for these variables in both datasets. [Table B.1](#) provides an example using a hypothetical stock, ‘XYZ,’ traded on exchange ‘ABC’ on a given trading day. In this case, three trades in the short sales file match exactly with three trades in the WCT file at 10:28:48 across all above-mentioned grouping variables. Consequently, these three trades are joined, and we assign the NBBO and LR trade classification from the WCT file to the corresponding short sales.

Step 2: Approximate matching

In the second step, we address cases where the number of trades in the short sales file is fewer than that in the WCT file with identical characteristics. For these instances, we match short sale trades to the first few corresponding WCT trades.

[Table B.2](#) illustrates this scenario. The short sales file contains two trades at 10:28:48, while the WCT file contains three trades at the same time. In this case, the two short sales are matched to the first two WCT trades. Although we cannot definitively identify which WCT trades are short sales, the NBBO information for these trades is likely consistent within this grouping, as per the NBBO construction methodology [\(Holden and Jacobsen, 2014\)](#).

Step 3: Residual matching

After applying steps 1 and 2, a very small subset of short sales remains unmatched. For these trades, we use a residual matching approach. Specifically, if no WCT trade has the exact timestamp as a short sale trade, we match the short sale to the prevailing NBBO quote in the WCT file, based on the timestamp.

[Table B.3](#) provides an example. A short sale occurs at 10:28:48, but no corresponding WCT trade is recorded at that time. In this case, we use the NBBO and LR classification from the most recent WCT NBBO quote (10:28:47).

Overall, this procedure allows us to merge quoted data into our short sales data and subsequently classify short sales into liquidity-supplying and liquidity-demanding trades. We are able to match 97.5% of the short sales to trades, comparable to the 98.2% match rate reported by [Comerton-Forde et al. \(2016\)](#). As in their study, we discard unmatched observations, which represent a small fraction of the sample.

Table A.1
Variable definitions.

Variable	Description
<i>ShortingFlow</i>	Shorting flow is the daily stock-level short volume divided by total trading volume across the three exchanges (NYSE, Nasdaq, and Cboe).
Liquidity-Supplying Shorting Volume, <i>LSS</i>	Short sales that take place above the prevailing midpoint are considered liquidity-supplying. <i>LSS</i> is defined as the daily sum of liquidity-supplying short volume divided by total trading volume across the three exchanges.
Liquidity-Demanding Shorting Volume, <i>LDS</i>	Short sales that take place below the prevailing midpoint are considered liquidity-demanding. <i>LDS</i> is defined as the daily sum of liquidity-demanding short volume divided by total trading volume across the three exchanges.
Liquidity-Supplying Long-Sales Volume, <i>LSSL</i>	Long sales that take place above the prevailing midpoint are considered liquidity-supplying. <i>LSSL</i> is defined as the daily sum of liquidity-supplying long-sales volume divided by total trading volume across the three exchanges.
Liquidity-Demanding Long-Sales Volume, <i>LDLS</i>	Long sales that take place below the prevailing midpoint are considered liquidity-demanding. <i>LDLS</i> is defined as the daily sum of liquidity-demanding long-sales volume divided by total trading volume across the three exchanges.
<i>ActiveUtilization</i>	The percentage of shares lent by custodians relative to the amount held by them in their lendable inventory pool. Data are obtained from Markit.
<i>ShortingFee</i>	Shorting Fee is proxied by the indicative fee, which is defined as the expected borrow cost for a hedge fund on a given day. Data are obtained from Markit.
Illiquidity, <i>ILQ</i>	The average ratio of the daily absolute return to the (dollar) trading volume over the past twelve months, following (Amihud, 2002). Data are obtained from www.openassetpricing.com .
Realized Spread, <i>RS</i>	Share weighted percent realized spread: $\sum_{k=1}^n w_k \frac{2D_k(P_k - M_{t+5})}{M_k}$, where M_{t+5} is the midpoint 5 min later after the k th trade and P_k is the transaction price. w_k is the share weight of each trade. D_k equals +1(−1) if trade k is a buy (sell) following the (Lee and Ready, 1991) classification algorithm. The variable is scaled by a factor of 10,000 to improve readability. Data are obtained from the 'Millisecond Intraday Indicators' by WRDS.
Price Impact, <i>PI</i>	Share weighted Percent Price Impact: $\sum_{k=1}^n w_k \frac{2D_k(M_{t+5} - M_k)}{M_k}$, where M_{t+5} is the midpoint 5 min later after the k th trade and M_k is the midpoint at the k th trade. The variable is scaled by a factor of 10,000 to improve readability. Data are obtained from the 'Millisecond Intraday Indicators' by WRDS.
<i>IntraVol</i>	Trade-based intraday volatility obtained from WRDS' 'Millisecond Intraday Indicators'. The variable is scaled by a factor of 10^6 to improve readability.
Order Imbalance, <i>OIB</i>	Total number of buys minus sells scaled by total number of transactions. Buy and sell volumes are classified by the algorithm of Lee and Ready (1991). Data are obtained from the 'Millisecond Intraday Indicators' by WRDS.
<i>Beta</i>	Coefficient of a 60-month rolling window regression of monthly stock returns minus the risk-free rate on market return minus the risk free rate. Data are obtained from www.openassetpricing.com .
<i>Size</i>	Market capitalization (stock price times shares outstanding, in \$ billion).
<i>BM</i>	Book-to-Market Ratio following (Fama and French, 1992).
<i>IdioVol</i>	Standard deviation of residuals obtained from Fama–French three factor regressions using the past month of daily data. Data are obtained from www.openassetpricing.com .
R_{t-1}	Stock return over the previous day.
$R_{t-5:t-2}$	Stock return over the previous week excluding the previous day.
Net Stock Issues, <i>NS</i>	Growth in number of split-adjusted shares between month $m - 18$ and $m - 6$. Data are obtained from www.openassetpricing.com .
Composite Stock Issues, <i>CS</i>	5-year growth in number of split-adjusted shares. Data are obtained from www.openassetpricing.com .
<i>Accruals</i>	Annual change in current total assets (act) minus annual change in cash and short-term investments (che) minus annual change in current liabilities (lct) minus annual change in debt in current liabilities (dlc) minus change in income taxes (txp). All divided by average total assets (at) over this year and last year. Data are obtained from www.openassetpricing.com .
Net Operating Assets, <i>NOA</i>	Difference between operating assets and operating liabilities, scaled by lagged total assets. Operating assets are total assets (at) minus cash- and short-term investments (che), operating liabilities are total assets minus long-term debt (dltt), minority interest (mib), deferred charges (dc) and book equity (ceq). Data are obtained from www.openassetpricing.com .
Asset Growth, <i>AG</i>	Annual growth rate of total assets (at). Data are obtained from www.openassetpricing.com .
Investments-to-Assets, <i>IA</i>	One-year change in property, plants and equipment (ppegt) plus one year change in inventory (inv), scaled by one-year lagged assets (at). Data are obtained from www.openassetpricing.com .
<i>Distress, Dist</i>	Distress is proxied by the Failure Probability of Campbell et al. (2008). Data are obtained from www.openassetpricing.com .
Momentum, $R_{m-12:m-2}$	Stock return between months $m - 12$ and $m - 2$.
Gross Profitability, <i>GP</i>	Revenue (sale) minus cost of goods solds (cogs), divided by 12 months lagged total assets (at). Data are obtained from www.openassetpricing.com .
Return on Assets, <i>ROA</i>	Quarterly net income (ibq) divided by lagged total assets (atq). Data are obtained from www.openassetpricing.com .

Table B.1
Exact matching.

Short sales				WCT						
Ticker	Time	Size	Price	Ticker	Time	Size	Price	NBB	NBO	LR
XYZ	10:28:25	100	10	XYZ	10:28:25	100	10	10	10.1	-1
⋮				XYZ	⋮					
XYZ	10:28:48	100	9.9	XYZ	10:28:48	200	9.9	9.9	10	-1
XYZ	10:28:48	100	9.9	XYZ	10:28:48	100	9.9	9.9	10	-1
XYZ	10:28:48	100	9.9	XYZ	10:28:48	100	9.9	9.9	10	-1
XYZ	10:28:50	200	9.9	XYZ	10:28:48	100	9.9	9.9	10	-1
⋮				XYZ	⋮					
XYZ	10:28:59	150	10.2	XYZ	10:28:59	150	10.2	10.1	10.2	+1

Table B.2
Approximate matching.

Short sales				WCT						
Ticker	Time	Size	Price	Ticker	Time	Size	Price	NBB	NBO	LR
XYZ	10:28:25	100	10	XYZ	10:28:25	100	10	10	10.1	-1
⋮				XYZ	⋮					
XYZ	10:28:48	100	9.9	XYZ	10:28:48	100	9.9	9.9	10	-1
XYZ	10:28:48	100	9.9	XYZ	10:28:48	100	9.9	9.9	10	-1
XYZ	10:28:50	100	10	XYZ	10:28:48	100	9.9	9.9	10	-1
⋮				XYZ	⋮					
XYZ	10:28:59	150	10.2	XYZ	10:28:59	150	10.2	10.1	10.2	+1

Table B.3
Residual matching.

Short sales				WCT						
Ticker	Time	Size	Price	Ticker	Time	Size	Price	NBB	NBO	LR
XYZ	10:28:25	100	10	XYZ	10:28:25	100	10	10	10.1	-1
⋮				XYZ	⋮					
XYZ	10:28:48	100	9.9	XYZ	10:28:45	100	9.9	9.9	10	-1
⋮				XYZ	10:28:46	100	9.9	9.9	10	-1
XYZ	10:28:59	150	10.2	XYZ	10:28:47	100	9.9	9.9	10	-1
⋮				XYZ	⋮					
XYZ	10:28:59	150	10.2	XYZ	10:28:59	150	10.2	10.1	10.2	+1

Appendix C. Price impact of liquidity-demanding and liquidity-supplying (short) sales

In this appendix, we complement our main findings with an intraday analysis in the spirit of [Comerton-Forde et al. \(2016\)](#). The purpose of this exercise is to examine whether the price impact patterns of short sales in our 2014–2018 sample resemble those documented by [Comerton-Forde et al. \(2016\)](#) for 2008. Although our main contribution focuses on longer-term return predictability, this analysis helps assess whether potential discrepancies with earlier work are due to differences in return horizons or evolving short-seller behavior over time.

We follow the procedure outlined in Section 2.2 and [Appendix B](#) to match each short sale transaction to its corresponding trade in the TAQ database using all available transaction details. Specifically, we employ the WCT database, which contains all equity trades alongside the best available NBBO quotes at the time of each trade. Using the [Lee and Ready \(1991\)](#) algorithm, we infer trade direction not only for all short sales but also for long sales across U.S. exchanges. This allows us to classify all trades into four distinct order flow categories: buyer-initiated short sales (liquidity-supplying short sales, *lss*), seller-initiated short sales (liquidity-demanding short sales, *lds*), buyer-initiated long sales (liquidity-supplying long sales, *lsls*), and seller-initiated long sales (liquidity-demanding long sales, *ldls*).²⁷ Our sample comprises over 17 billion trades (compared to about 273 million trades in [Comerton-Forde et al. \(2016\)](#)).

²⁷ To distinguish from the daily relative trading volume measures in the main text, we use lowercase notation for the order flow variables in this section.

Following [Comerton-Forde et al. \(2016, Table 6\)](#), we estimate each order type's average price impact. More specifically, for a given stock *i* and post-trade horizon *k*, we regress cumulative market-adjusted midpoint returns $r_{i,t:t+k}$ on indicator variables for *ldls*, *lss*, and *lsls* trades:

$$r_{i,t:t+k} = \mu_i + \gamma_t + \beta_1 \mathbb{1}\{ldls_{i,t}\} + \beta_2 \mathbb{1}\{lss_{i,t}\} + \beta_3 \mathbb{1}\{lsls_{i,t}\} + \varepsilon_{i,t:t+k}, \quad (\text{C.1})$$

where liquidity-demanding short sales, *lds*, form the omitted category, and the coefficients measure the incremental price impact relative to *lds*. We estimate three versions of this regression: Model 1 without fixed effects, Model 2 with stock fixed effects, and Model 3 with five-minute interval fixed effects. Standard errors are clustered by stock and date. We consider price impact horizons of five minutes, 30 min, two hours, and 24 h. Each trade is weighted by its dollar volume.²⁸

Following [Comerton-Forde et al. \(2016\)](#), we group all trades into fixed five-minute intervals, and returns for each trade are computed based on the midpoint return over the corresponding interval, regardless of the exact timestamp of the trade within that interval. For example, any trade occurring between 11:30:00 and 11:34:59 is assigned the five-minute return from 11:30 to 11:35. As a result, all trades within a given five-minute window share the same returns.

We present the estimation results in [Table C.1](#). We generally confirm the findings of [Comerton-Forde et al. \(2016\)](#). First, liquidity-demanding sales (both short and long) are associated with a negative price impact, whereas liquidity-supplying sales exhibit a positive price impact. In particular, short sales that demand liquidity, *lds*, the omitted category,

²⁸ We also estimate the models where each trade is equal-weighted. The unreported results for this alternative weighting method are very similar and available upon request.

Table C.1
Price impact measured through varying post-trade horizons.

		5 minutes			30 minutes			2 h			24 h		
		Model 1	Model 2	Model 3	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
Panel A: Pooled													
Intercept		−2.29	−2.19	−2.30	−2.31	−2.22	−2.27	−2.60	−2.48	−2.50	−3.05	−2.89	−2.82
		(−25.68)	(−28.24)	(−23.69)	(−17.72)	(−21.71)	(−19.36)	(−9.95)	(−14.19)	(−13.11)	(−5.48)	(−10.25)	(−7.08)
	$\mathbb{1}\{ldls\}$	−0.58	−0.61	−0.57	−0.18	−0.22	−0.26	0.45	0.35	0.26	1.48	1.31	1.03
		(−7.58)	(−10.05)	(−7.39)	(−1.50)	(−2.26)	(−2.30)	(2.17)	(2.22)	(1.52)	(4.00)	(4.47)	(3.63)
$\mathbb{1}\{lss\}$		5.07	4.91	5.07	4.98	4.82	4.99	5.03	4.85	5.06	5.03	4.83	5.12
		(28.06)	(29.10)	(28.09)	(25.85)	(27.16)	(25.92)	(24.03)	(25.53)	(24.22)	(17.95)	(19.27)	(18.48)
$\mathbb{1}\{lsls\}$		5.52	5.31	5.54	5.85	5.69	5.77	6.54	6.33	6.33	7.87	7.59	7.37
		(25.25)	(26.44)	(25.49)	(24.58)	(26.68)	(24.56)	(24.27)	(26.98)	(24.08)	(19.72)	(22.30)	(21.12)
Panel B: By size quintile													
Q1 Intercept		−12.97	−11.74	−13.06	−13.12	−12.09	−13.27	−15.52	−14.13	−15.57	−21.06	−18.52	−21.05
		(−36.77)	(−46.79)	(−38.08)	(−23.20)	(−39.25)	(−20.46)	(−11.22)	(−37.56)	(−12.14)	(−5.56)	(−20.90)	(−6.09)
	$\mathbb{1}\{ldls\}$	−2.92	−2.77	−2.83	−3.13	−3.21	−3.27	−1.48	−1.99	−1.60	4.48	1.70	4.16
		(−8.28)	(−14.06)	(−8.05)	(−6.76)	(−8.82)	(−6.59)	(−3.22)	(−3.42)	(−2.68)	(2.20)	(1.11)	(1.77)
$\mathbb{1}\{lss\}$		26.69	24.09	26.65	27.95	25.22	27.90	28.13	25.13	27.97	29.85	25.43	29.38
		(48.08)	(57.33)	(47.84)	(43.65)	(50.40)	(42.84)	(34.43)	(39.09)	(33.27)	(12.77)	(19.37)	(12.59)
$\mathbb{1}\{lsls\}$		30.33	27.82	30.64	31.26	29.87	32.03	33.05	31.00	33.52	36.87	34.03	37.56
		(30.28)	(51.74)	(35.35)	(13.56)	(46.84)	(22.12)	(15.69)	(39.91)	(22.79)	(7.95)	(21.06)	(10.23)
Q2 Intercept		−7.82	−6.95	−7.83	−7.96	−7.02	−7.96	−7.86	−6.83	−7.71	−9.15	−8.03	−8.91
		(−35.22)	(−47.73)	(−39.24)	(−17.79)	(−33.82)	(−24.64)	(−7.00)	(−14.09)	(−10.23)	(−3.73)	(−9.49)	(−4.92)
	$\mathbb{1}\{ldls\}$	−2.12	−2.26	−2.08	−2.46	−2.64	−2.43	−1.30	−1.65	−1.47	1.17	0.59	0.93
		(−11.06)	(−15.82)	(−10.72)	(−7.76)	(−9.95)	(−7.69)	(−2.37)	(−3.51)	(−2.41)	(1.13)	(0.56)	(0.91)
$\mathbb{1}\{lss\}$		17.02	15.59	17.00	16.88	15.41	16.91	16.30	14.83	16.33	15.80	14.32	15.90
		(56.19)	(62.01)	(56.40)	(51.17)	(52.68)	(51.94)	(37.81)	(38.45)	(38.15)	(21.39)	(22.09)	(21.38)
$\mathbb{1}\{lsls\}$		20.04	18.15	20.05	20.87	18.80	20.79	21.80	19.54	21.36	23.52	21.14	22.70
		(39.87)	(51.06)	(40.64)	(37.25)	(46.37)	(39.25)	(30.55)	(36.03)	(31.95)	(18.90)	(18.04)	(18.66)
Q3 Intercept		−5.23	−4.68	−5.23	−5.55	−4.93	−5.56	−6.48	−5.79	−6.54	−8.29	−7.71	−8.47
		(−24.43)	(−34.92)	(−29.76)	(−15.83)	(−27.86)	(−20.41)	(−8.31)	(−17.71)	(−10.66)	(−4.59)	(−12.05)	(−5.77)
	$\mathbb{1}\{ldls\}$	−1.31	−1.46	−1.31	−0.96	−1.20	−0.93	−0.04	−0.37	0.11	1.31	1.10	1.76
		(−7.39)	(−10.18)	(−7.30)	(−4.32)	(−6.31)	(−4.38)	(−0.11)	(−1.15)	(0.32)	(1.96)	(1.87)	(3.06)
$\mathbb{1}\{lss\}$		11.35	10.43	11.35	11.28	10.32	11.30	11.02	10.02	11.09	10.09	9.19	10.27
		(42.05)	(48.98)	(41.83)	(39.30)	(44.18)	(39.24)	(32.47)	(35.35)	(32.96)	(21.37)	(21.82)	(22.53)
$\mathbb{1}\{lsls\}$		12.98	11.85	12.98	13.69	12.42	13.68	14.70	13.27	14.73	15.77	14.53	15.88
		(34.91)	(39.14)	(34.99)	(34.33)	(38.14)	(34.36)	(29.59)	(32.52)	(29.97)	(23.51)	(24.89)	(24.19)
Q4 Intercept		−3.20	−2.93	−3.20	−3.28	−2.97	−3.26	−4.32	−3.96	−4.28	−5.69	−4.94	−5.57
		(−28.10)	(−37.46)	(−30.12)	(−12.41)	(−23.87)	(−15.91)	(−7.63)	(−19.48)	(−9.55)	(−4.40)	(−15.36)	(−5.17)
	$\mathbb{1}\{ldls\}$	−0.79	−0.86	−0.79	−0.38	−0.52	−0.43	0.39	0.21	0.30	1.84	1.87	1.59
		(−11.50)	(−12.90)	(−11.54)	(−2.27)	(−3.45)	(−2.84)	(1.46)	(0.81)	(1.28)	(3.72)	(4.68)	(3.71)
$\mathbb{1}\{lss\}$		7.07	6.64	7.07	6.89	6.43	6.92	6.95	6.48	7.01	6.85	6.01	7.01
		(40.34)	(48.76)	(40.29)	(33.50)	(40.79)	(33.63)	(29.30)	(32.81)	(30.29)	(17.25)	(21.74)	(18.44)
$\mathbb{1}\{lsls\}$		7.78	7.22	7.79	8.23	7.58	8.17	9.16	8.41	9.04	10.57	9.30	10.19
		(31.27)	(36.34)	(30.95)	(26.50)	(31.81)	(27.05)	(27.43)	(29.64)	(29.18)	(22.32)	(23.86)	(24.22)
Q5 Intercept		−1.51	−1.42	−1.52	−1.48	−1.42	−1.44	−1.55	−1.47	−1.46	−1.60	−1.47	−1.39
		(−15.58)	(−19.48)	(−14.95)	(−11.93)	(−15.57)	(−12.61)	(−6.57)	(−8.51)	(−9.67)	(−3.40)	(−5.00)	(−5.98)
	$\mathbb{1}\{ldls\}$	−0.40	−0.44	−0.38	0.03	0.01	−0.03	0.57	0.48	0.40	1.35	1.23	0.95
		(−4.88)	(−8.30)	(−4.93)	(0.27)	(0.10)	(−0.25)	(2.48)	(2.93)	(2.12)	(3.22)	(3.88)	(2.96)
$\mathbb{1}\{lss\}$		3.32	3.18	3.32	3.22	3.11	3.23	3.34	3.21	3.35	3.47	3.30	3.51
		(19.48)	(22.72)	(19.52)	(16.93)	(19.77)	(17.08)	(14.80)	(17.51)	(15.08)	(10.32)	(11.95)	(10.80)
$\mathbb{1}\{lsls\}$		3.63	3.45	3.65	3.88	3.77	3.81	4.46	4.33	4.27	5.71	5.50	5.28
		(17.13)	(20.41)	(17.06)	(16.74)	(20.28)	(16.67)	(16.44)	(20.39)	(16.61)	(12.97)	(15.87)	(14.02)

This table reports regression estimates in which the dependent variable is the market-adjusted midquote price impact measured in bps over future horizons ranging from 5 min to 24 h. $\mathbb{1}\{ldls\}$, $\mathbb{1}\{lss\}$, and $\mathbb{1}\{lsls\}$ are indicator variables that take the value of one for seller-initiated long sales (liquidity-demanding long sales), buyer-initiated short sales (liquidity-supplying short sales), and buyer-initiated long sales (liquidity-supplying long sales), respectively, and zero otherwise. Liquidity-demanding short sales form the omitted category. Panel A shows results for the pooled sample and Panel B shows results for each of the five size quintiles based on the previous month's stock market capitalization distribution of the sample. Model 1 does not have fixed effects, Model 2 includes fixed effects on stocks and Model 3 includes fixed effects on five-minute intervals. Each trade is weighted by dollar volume. Standard errors are clustered by stock and by date. The sample includes 3,248 stocks and 17,261,228,313 trades from 2014 through 2018.

display an initial 5 min price impact of −2.29 bps, which increases in magnitude to −3.05 bps over a 24-h horizon. These magnitudes are similar to the estimates reported by [Comerton-Forde et al. \(2016\)](#), who document impacts of −2.26 and −3.46 bps, respectively, and thus a somewhat larger price drift over the next 24 h.

Liquidity-demanding long sales, *ldls*, exhibit a stronger impact of −2.87 bps in the first five minutes, but this effect partially reverts over time, decreasing in magnitude to −1.57 bps over the next 24 h. Even though the stronger initial price impact of long sales might appear

at odds with the idea that short sellers are particularly informed, the reversion in returns is consistent with the findings of [Comerton-Forde et al. \(2016\)](#), and suggests that these long sales are less informed and are more likely to reflect transient imbalances.

Turning to liquidity-supplying short sales, we find that *lss* is associated with a return of 2.78 (= −2.29 + 5.07) bps over the first five minutes. After this initial positive price impact, liquidity-supplying short sales exhibit a negative drift yielding a cumulative return of 1.98 (= −3.05 + 5.03) bps over 24 h. For the same trade type, the

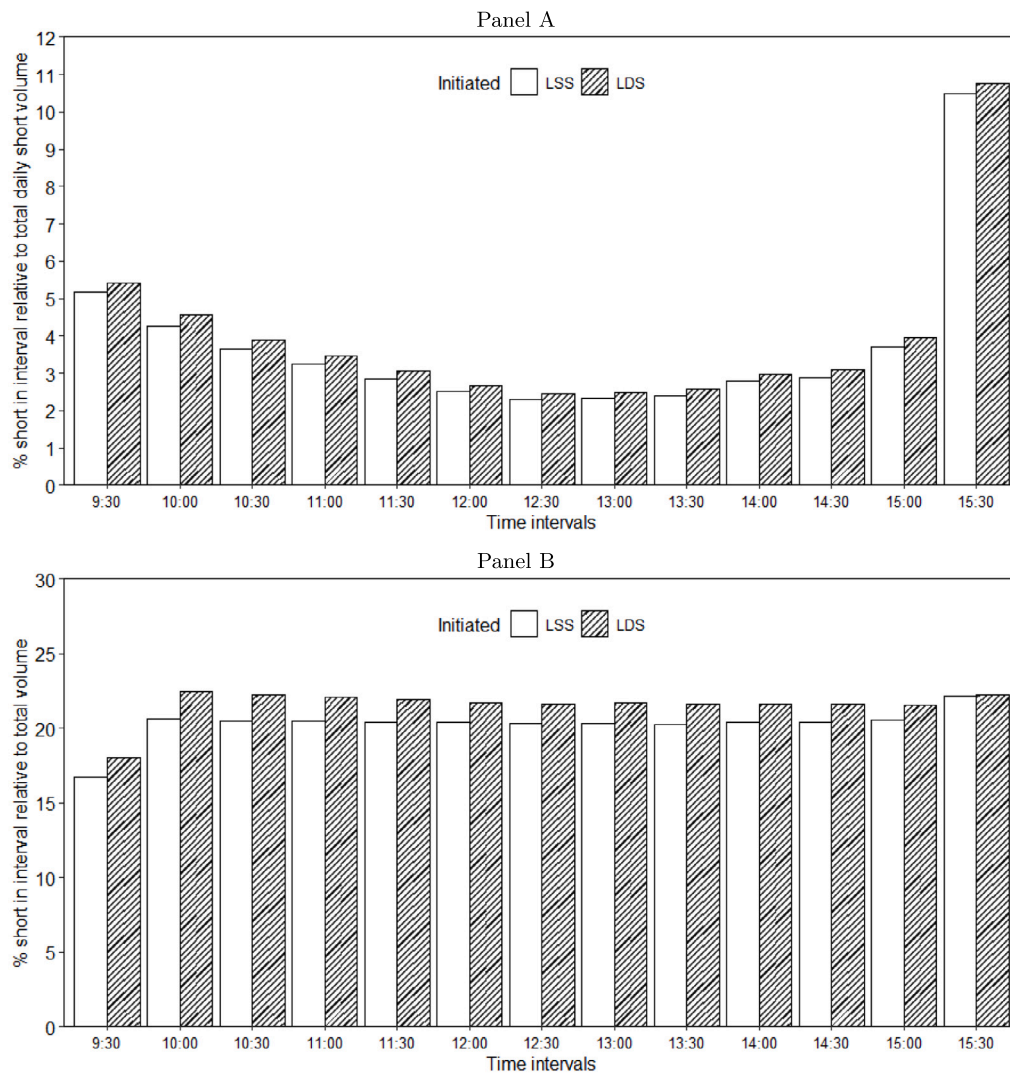


Fig. D.1. Intraday short volume.

This figure shows the percentage of liquidity-supplying (*LSS*) and liquidity-demanding (*LDS*) short-selling volume within 30 min intervals as a proportion of the total daily short selling volume (Panel A) and as a proportion of the trading volume within the same 30 min interval (Panel B). Striped bars indicate *LSS*, while white bars represent the volume of *LDS*. The sample period is 2014 through 2018.

corresponding values in [Comerton-Forde et al. \(2016\)](#) estimations are 2.63 and 2.09 bps. These estimates suggest a similar intraday pattern, although the negative drift for *lss* is somewhat larger for our recent sample period.

Lastly, liquidity-supplying long sales, *lsls*, are also associated with a positive price impact, which persists or even increases slightly over the 24-h horizon, consistent with uninformed trading and [Comerton-Forde et al. \(2016\)](#) findings.

We also follow [Comerton-Forde et al. \(2016\)](#) in estimating these regressions across quintiles of the market capitalization distribution. We find that all patterns discussed above hold across size bins, although the price impact is larger for smaller stocks compared to [Comerton-Forde et al. \(2016\)](#) estimates. This is to be expected due to differences in sample composition. [Comerton-Forde et al. \(2016\)](#) focus on 350 NYSE-listed stocks, which tend to have a larger market capitalization compared to stocks listed on other exchanges, whereas our sample covers the full universe of U.S. equities. In other words, our lowest quintile sample is composed of smaller stocks compared to their corresponding quintile and, therefore, shows a larger price impact for both seller- and buyer-initiated trades.

Taken together, these results show that both liquidity-demanding and liquidity-supplying short sellers in our sample exhibit similar price impact patterns in the 24-h post-trade period. At the same time, the negative price drifts after the initial five-minute price impact are similar for the two types of short sales in our sample. This similarity is consistent with the one-day ahead return predictability based on *LSS* and *LDS* depicted in [Fig. 2](#). Thus, the apparent contrast between our long-horizon findings and the intraday findings in [Comerton-Forde et al. \(2016\)](#) is likely driven by the return horizon difference that is one of the key ways in which our paper's primary focus differs from that of [Comerton-Forde et al. \(2016\)](#).

Appendix D. Additional figures

See [Figs. D.1 and D.2](#).

Data availability

[Stealthy Shorts: Informed Liquidity Supply \(Reference data\)](#) (Mendeley Data)

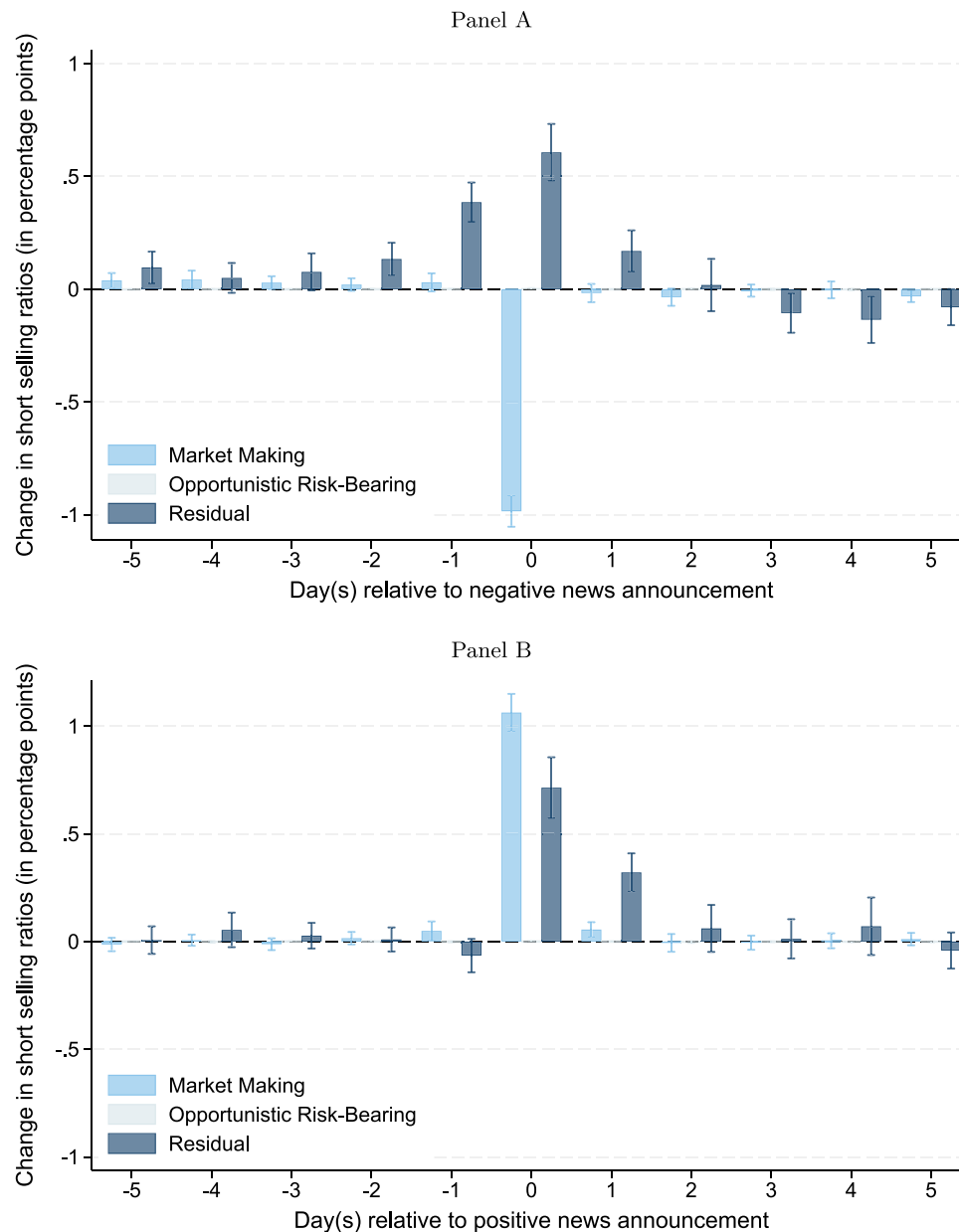


Fig. D.2. Components of LSS around news event days including risk-bearing component.

This figure shows the change of the residual, the market-making component, and the risk-bearing component of LSS in percentage points around firm-specific news events. Each bar is based on a panel regression with stock and day fixed effects. The dependent variable is one of the daily components of LSS and the independent variable of interest is an indicator variable that takes the value one if a news story occurs and zero otherwise. In each regression we vary the timing of the dependent variable relative to the news event to examine short-volume changes around news. For example, $t - 1$ indicates that the dependent variable is observed one day prior to the news event. To control for the response of short sellers to past returns, we include five-day lags of daily returns (where the lags are relative to the timing of the dependent variable). In Panel A and Panel B we examine changes in LSS components around negative and positive news events, respectively. We define a news event as negative (positive) if the announcement day return is in the bottom (top) quintile of returns on a given day, respectively. The sample period is 2014 through 2018.

References

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